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THESIS

Spontaneous deamination of cytosine in DNA – modelling the impact of nucleotide sequence

Spontaniczna deaminacja cytozyny w DNA – modelowanie wpływu sekwencji nukleotydowej

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cytosine deamination, 5-methylcytosine, C>T transition mutations, DNA sequence context; transition state stabilization, MED model, DTSS

abstract

Spontaneous deamination of cytosine and 5-methylcytosine is an important source of C>T transition mutations in DNA, especially in methylated sequence contexts. This thesis investigates how neighboring nucleotides affect this process using MED and DTSS approaches. The results identify sequence contexts that may stabilize the transition state and promote deamination

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1. Theoretical background

1.1. Biological overiew

Deamination is a spontaneous chemical reaction that occurs in an aqueous environment, in which a nitrogenous base loses its exocyclic amino group. This change modifies its hydrogen bonding properties, causing it to no longer recognize its typical partner and allowing it to pair with another nucleotide. As a result, incorrect base pairs may appear in the DNA [1]. In the case of cytosine (C), the product of deamination is uracil (U) (Figure 1), while for 5-methylcytosine (5-mC), the product is thymine (T) (Figure 2) [1-3].

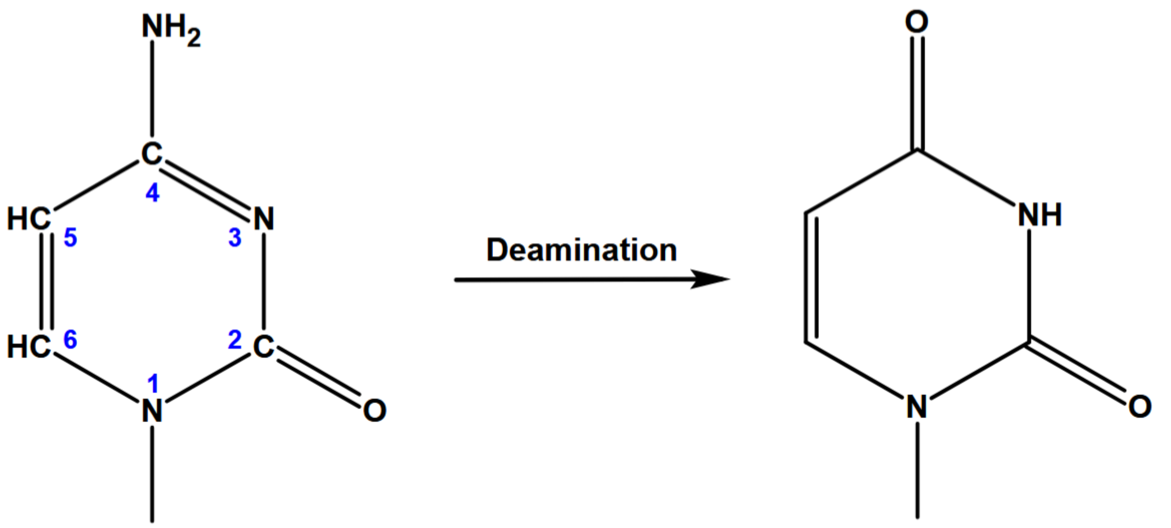

Fig. 1 Deamination of cytosine to uracil.

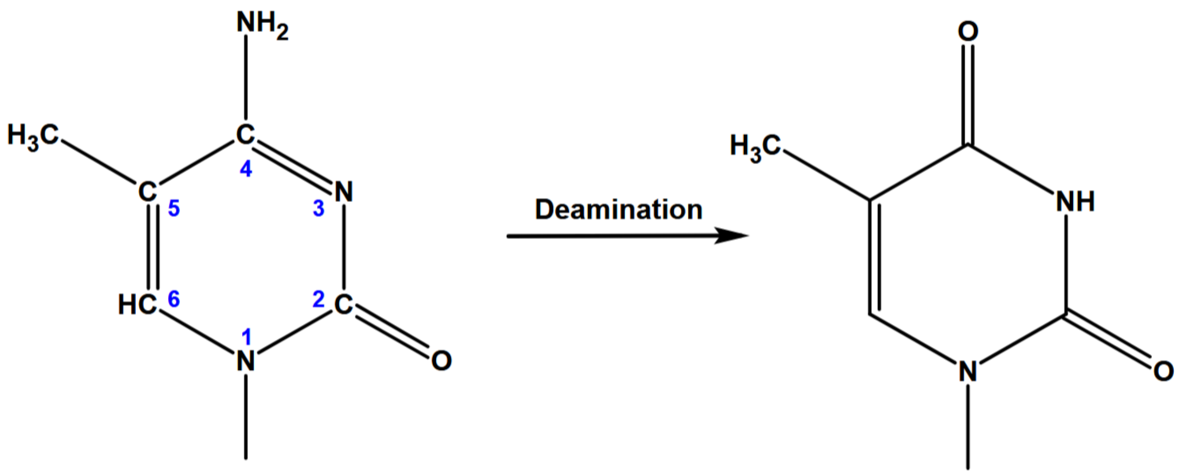

Fig. 2 Deamination of 5-methylocytosine to thymine.

Both reactions belong to a broader group of spontaneous processes that threaten DNA integrity, such as hydrolysis and oxidation, and are associated with mutation accumulation and cancer-related processes [1]. Importantly, the quantitative burden can be substantial in mammals: cytosine deamination is estimated to generate approximately 100-500 uracil residues per mammalian genome per day, indicating that U:G mismatches occur frequently and pose a constant challenge to DNA maintenance and repair pathways [1].

This change causes a mismatch in the double-stranded DNA. In a normal duplex, cytosine is paired with guanine, but after deamination, uracil appears instead of cytosine, resulting in a U:G mismatch. If such damage is not repaired before replication, uracil can be read as thymine and serve as a template for adenine insertion. As a result, after the next replication cycle, the original G:C pair may be fixed as A:T, i.e. a stable change of the C:G base pair to T:A in the genome [4].

A similar mechanism applies to methylated CpG sites, where CpG refers to a cytosine followed by a guanine in the 5′→3′ direction of the same DNA strand, linked by a phosphate group [5], [6]. Deamination of 5-methylcytosine, which is also originally paired with guanine, leads to the formation of thymine and T:G mismatch. Because thymine is a normal DNA base, the lesion is not flagged by the presence of an “incorrect” base and must be detected solely through mispairing, which can delay or reduce the efficiency of repair [1]. If T:G is not repaired before replication, the likelihood of C→T transition being perpetuated increases after subsequent copying cycles [4].

The CpG dinucleotide is particularly important because it is the main target of cytosine methylation in vertebrate genomes. In these sites, cytosine is often methylated at the C-5 position, leading to the formation of 5-mC. Importantly, 5-mC residues act as mutation hotspots, as methylated cytosines are more prone to spontaneous deamination. Experimental observations show that sequences containing 5-mC have an increased frequency of G:C→A:T transitions, which is consistent with the more frequent formation of T:G mismatches after 5-mC deamination [7]. Early studies provided direct evidence that deamination of 5-mC to thymine contributes to spontaneous mutation, supporting the view that cytosine methylation can be intrinsically mutagenic [7]. Subsequent analyses of transition mutations in vertebrate genomes further demonstrated a strong association between CpG methylation and elevated substitution rates, placing methylated CpG sites at the core of a major endogenous mutational pathway [4].

This mutagenic pathway is also visible at the whole-genome level as a repetitive mutation pattern. In large cancer genome-wide datasets, the signature of the single base substitution SBS1 is dominated by C>T substitutions in CpG dinucleotides and attributed to spontaneous methylcytosine deamination, demonstrating on a population scale that 5-mC deamination is a common endogenous source of mutations [8].

If such a mismatch is not corrected, it may be perpetuated as a C→T transition on the strand containing the original cytosine, and equivalently described as a CpG→TpG or CpG→CpA change on the complementary strand. Over a longer period of time, repeated cycles of deamination and incomplete repair lead to a gradual depletion of CpG motifs in the genome and an increase in the frequency of TpG and CpA dinucleotides [4,9].

Because CpG-associated C→T transitions are so frequent, they contribute significantly to the mutation burdens observed in human diseases, including cancers and hereditary disorders. Spontaneous chemical reactions occurring in DNA, including base deamination, are considered one of the main sources of mutations that can lead to cell death and promote cancerous processes [1]. C→T transitions within CpG dinucleotides are particularly important because methylated CpGs are mutation hotspots. In quantitative terms, it is estimated that CpG sites may account for approximately 30% of point mutations in the germline, and in somatic mutations in cancers, a classic example is methylated CpGs in the

TP53 gene, which may account for up to 50% of all inactivating mutations in colorectal cancer [10].

Cytosine methylation disorders and the resulting loss of epigenetic control also play a significant role in the etiology of many neurodevelopmental syndromes. In fragile X syndrome, the expansion of CGG repeats leads to abnormal de novo methylation and silencing of FMR1 gene transcription. In ICF syndrome, mutations in DNMT3B are associated with hypomethylation of centromeric regions and chromosome destabilization. Rett syndrome is particularly relevant in the context of mutagenesis, as mutation hotspots in CpG involving C→T transitions have been described in the MECP2 gene. Reports also indicate that methylation defects may occur in ATR-X syndrome, further emphasizing the importance of proper epigenetic regulation in the development of the nervous system. From the perspective of this study, these examples show that the processes occurring at 5-mC in CpG sites are important for the regulation of gene activity, but also as a potential source of permanent changes in DNA sequences [10].

The persistence or removal of deamination products in the genome largely depends on the efficiency of base excision repair (BER). The BER pathway removes single, endogenous base damages before they develop into permanent sequence changes. Its key step is initiation by DNA glycosylase, which removes the damaged base by cleaving the N-glycosidic bond and creating an AP site, followed by AP endonuclease cleaving the strand and DNA polymerase and ligase restoring sequence continuity. In the case of uracil in a U:G pair, the principal enzyme responsible for its removal is uracil-DNA glycosylase (UNG), while single-strand-selective monofunctional uracil-DNA glycosylase 1 (SMUG1), another DNA glycosylase enzyme, can serve as a backup and partially compensate when UNG activity is reduced [1,11].

When both UNG and SMUG1 are impaired, the number of mutations increases significantly, including a higher frequency of G:C→A:T transitions. This suggests that both enzymes operate in two partially independent mechanisms of genome protection, so the loss of both at once has a stronger effect than the loss of only one of them [1]. Correcting T:G mismatches after 5mC deamination is much more difficult because the reaction product is a ordinary DNA base, so the damage must be detected solely as a mismatch. In mammalian cells, this process relies more on DNA glycosylases such as thymine-DNA glycosylase (TDG) and methyl-CpG-binding domain protein 4 (MBD4), which act on double-stranded DNA and display relatively low catalytic efficiency, making G:T repair less effective than the rapid removal of G:U mismatches [1,4].

Although the effects of cytosine and 5-mC deamination are usually described in terms of base mismatches and repair pathway activity, they are fundamentally based on a specific hydrolytic reaction involving the conversion of an amino group into a carbonyl group. The propensity of a given site to convert C→U or 5 mC→T depends on the course of the reaction at the level of elementary steps, such as the activation of a water molecule as a nucleophile, the reorganization of hydrogen bonds, and coupled proton transfers leading to a transition state. The rate of these steps is influenced by the local microenvironment of the DNA [4,12].

1.2. Chemistry and reaction mechanism

Hydrolytic deamination of cytosine and 5-methylcytosine in solution involves the substitution of an exocyclic amino group by a carbonyl group, leading to uracil (from C) or thymine (from 5-mC), respectively. First, a nucleophile derived from water initiates the reaction, a transient tetrahedral structure is formed, proton transfers reorganize the hydrogen bond network, and the amino group is eliminated as NH_3 with the restoration of the carbonyl group. An essential requirement for the elimination step is proton rearrangement that converts the amino substituent into a leaving group, since only a protonated amino group can depart efficiently. In uncatalyzed settings this protonation is achieved by the surrounding water network rather than by a dedicated catalytic acid, and the availability of water-mediated proton relays can therefore determine which microscopic pathway is kinetically accessible and whether carbonyl restoration can proceed [13,14].

The same complete conversion can occur as an enzymatically catalyzed reaction in metabolic pathways or as an uncatalyzed process in an aqueous environment and in DNA, where it is a permanent source of spontaneous base damage. In practice, the reaction involves several coupled steps of protonation and deprotonation, and their course may depend on the conditions. Deamination is particularly sensitive to solvent composition, pH, temperature, and, in DNA, to the local microenvironment created by neighboring bases and hydration patterns [15].

In the literature, non-enzymatic hydrolytic deamination of cytosine is commonly described as an addition-elimination reaction, the details of which depend on solution conditions and on the protonation state of the base. For acidic and near-neutral environments, a frequently discussed stepwise scenario is initiated by protonation at N3, which increases susceptibility to nucleophilic addition and enables a sequence of elementary steps ending with deamination at N4 (departure of the amino fragment and formation of the carbonyl) [2]. Computational models further suggest that, even with the same overall outcome, the uncatalyzed process can proceed along multiple elementary routes. Because the detailed sequence of proton transfers and the arrangement of water molecules are largely reconstructed on the basis of calculations, the mechanism is best viewed as a family of closely related micro pathways that realize the same general transformation under different local solvation patterns [14,16].

The influence of pH follows from the coupling between nucleophile generation and proton-transfer equilibria. Neutral water is a weak nucleophile and is generally insufficient to initiate direct addition with appreciable probability, whereas hydroxide has sufficient nucleophilicity to promote attack . In stepwise descriptions, a key enabling event can be transient protonation at N3, which can generate a short-lived protonated base together with an OH^- equivalent that supports nucleophilic addition and the formation of a tetrahedral intermediate. This addition step disrupts aromaticity and often accounts for a substantial fraction of the activation barrier, because it couples bond formation with electronic reorganization within the ring and partial charge development on heteroatoms. Temperature further modulates the observed rate by increasing the probability of surmounting the barrier and by altering solvent dynamics, which affects how efficiently hydrogen-bond networks reorganize along the reaction coordinate [15,17].

Because water is not only a solvent but can participate directly in the reaction coordinate, explicit multi-water ("water-assisted") descriptions are frequently used. Such models incorporate additional water molecules that reorganize hydrogen bonds, donate and accept protons, and stabilize transition structures, typically lowering activation barriers relative to minimal descriptions [14,15]. In more explicit cluster models with two and three water molecules, multi-step pathways with coupled proton relays have been identified. In the three-water variant, a late proton transfer to the amino nitrogen, correlated with NH_3 formation and release, has been proposed as a rate-limiting event, and the presence of additional water molecules can stabilize intermediates and reduce barriers relative to models with fewer waters [13]. Across these variants, the main differences lie in how the proton relay is organized and when bond reorganization occurs, while the overall chemical outcome remains the same [13,14]. Alternative proposals that place C4–N4 cleavage earlier and proceed through a distinct tautomeric intermediate have also been discussed, but they are generally less favorable energetically [14].

The above reaction landscape motivates treating deamination as a family of related micro pathways that perform the same transformation in different local solvation environments, which is also useful in sequence-dependent analyses (e.g., by comparing transition-state stabilization relative to the substrate). In comparative sequence studies, these effects are often quantified using $\Delta\Delta E^{\ddagger}$, which captures how changes in neighboring nucleotides alter the stabilization of the transition state relative to the substrate for the same underlying transformation [14].

Earlier proposals suggested that CpG methyltransferases might facilitate deamination as a side reaction. However, experimental tests did not support increased C→U formation in double-stranded DNA, which points to spontaneous hydrolysis as the primary source of CpG-associated transitions [4]. Under near-physiological conditions, deamination at G-5-mC proceeds substantially faster than at G-C, providing a kinetic benchmark that highlights an intrinsic chemical difference between the two base pairs. This contrast is useful when interpreting calculated activation-barrier shifts ($\Delta\Delta E^{\ddagger}$), as it illustrates how relatively small energetic differences can translate into measurable changes in hydrolysis rates across sequence contexts [4].

Enzymatic deamination provides a benchmark in this context. More generally, enzyme-based analyses highlight two requirements that also apply to spontaneous deamination. Generating a sufficiently reactive nucleophile and ensuring protonation of the departing amino group so that it can leave as NH_3. Enzymes fulfil both by activating bound water, reorganizing substrate geometry and stabilizing charge development along the reaction coordinate through specific electrostatics and hydrogen-bonding interactions [17]. It retains the same chemical core but imposes an ordered microenvironment that lowers the effective barrier and narrows the set of accessible pathways. In zinc-dependent cytosine deaminases, Zn^{2+} activates water at the active site and stabilizes charges, while proton transfers are guided by conserved acidic residues. For bacterial cytosine deaminase, the roles of Glu217 and Asp313 in nucleophile activation and substrate protonation have been described, promoting efficient NH_3 release [17]. A similar picture has been obtained for yeast cytosine deaminase, where controlled stages of ammonia release and subsequent uracil departure from the active site are important [18]. An example of deamination occurring directly in DNA are enzymes from the APOBEC family, which catalyze the conversion of C→U in single-stranded DNA, and APOBEC3A has also been reported to efficiently deaminate 5-methylcytosine [19].

The reactivity of cytosine and 5-methylcytosine is also influenced by tautomers and protonation states, and both of these phenomena may differ between these nitrogen bases. Cytosine can exist in several tautomer forms, including amino-oxo (C1), imino (C2) and amino-hydroxyl (C3), which are energetically close, and their relative stability may depend on the calculation method used [20]. Despite this energetic 'proximity', the amino form ($-NH_2$) dominates in DNA, while imino forms (=NH) are rare [3]. Their significance stems from the fact that even a short-lived appearance of a rare tautomer changes the arrangement of hydrogen bond donors and acceptors, which may promote unusual base pairs and errors during replication, followed by the fixation of mutations [3,20]. More recent analyses additionally indicate that rare imino tautomers may form as neutral proton transfer products (described as PC), which increases their stability relative to charged variants and is considered one of the causes of spontaneous mutations in G–C pairs [21].

Protonation is also important because it strongly modulates the reactivity of the pyrimidine ring. In hydrolytic deamination mechanisms, protonation at the N3 position is particularly common as a step initiating or facilitating the reaction in acidic and near-neutral environments, leading to an addition-elimination scenario [20]. The result is protonated forms, which are much stronger electrophiles than neutral forms. Protonation can therefore significantly facilitate nucleophilic attack by lowering the reaction barrier through stabilization of the transition state. This example shows that methylation at the C5 position affects the protonation (acid-base) equilibrium of the base [13].

It has also been shown that 5-methylation increases the pK of cytosine (from approximately 4.45 to 4.6), which increases the fraction of protonated forms at physiological pH and may promote greater susceptibility to hydrolytic deamination [4,22]. Ultimately, this means that the local microenvironment of DNA can simultaneously modulate the frequency of rare tautomers (affecting pairing fidelity) and the availability of protonated forms (affecting reactivity), and CpG methylation can further shift these equilibria towards more ‘chemically active’ states. [4,20,21].

1.3. Sequence dependence of DNA reactivity

Reactivity of cytosine and 5-methylcytosine in DNA depends not only on the base itself, but also on which nucleotides are directly adjacent to it in the sequence. In double-stranded DNA, their hydrolytic deamination does not take place in bulk water; instead, it occurs within a structured local microenvironment defined by base stacking, the electrostatic field of the phosphate backbone, and groove hydration patterns, all of which can modulate how readily the reaction proceeds [23,24]. The neighboring nucleotides on the 5′ and 3′ sides therefore shape the local thermodynamic and structural stability within the DNA helix [23]. Because the activation barrier of cytosine deamination is sensitive to local solvation, proton-transfer steps, and the arrangement of hydrogen-bond networks, variations in the local reaction environment may influence the reaction rate [24]. As a result, even at the same pH and temperature, deamination can be faster in some DNA sequences than in others because the neighboring bases create different local environments around the reacting site [6,25,27].

A first source of sequence effects is base stacking and the associated local electrostatic field. Nearest-neighbor interactions modulate the geometry and thermodynamic stability of local base-pair, which influences the accessibility of reactive conformations of the target cytosine or 5-mC. At the same time, sequence-dependent differences in stacking geometry and in the distribution of partial charges can either stabilize or destabilize the more polarized electronic structure associated with the transition state, shifting the balance of stabilization between the substrate and the transition state. In this way, local electrostatics can speed up or slow down the reaction by selectively stabilizing (or destabilizing) the transition state, even though the overall chemical transformation remains the same [26].

A second major determinant of the deamination process is the direct participation of water molecules via hydrogen-bond networks [14]. Additional water molecules can act as a bridge to facilitate proton transfer (a water-assisted mechanism), effectively stabilizing the transition state and lowering the activation free energy barrier for the underlying chemical transformation [14,15,28,29]. Separately, at the macromolecular level, sequence-dependent DNA geometry shapes the local hydration patterns in the minor groove, dictating the population and organization of ordered water molecules [27]. Because reaching the transition state requires solvent reorganization and charge redistribution, the presence or absence of a stable hydrogen-bond network involving water molecules and functional groups of nearby bases can substantially modify the energetic cost of this step. In the context of deamination, potential proton-transfer pathways may further contribute, including transient "proton wires" formed by water molecules together with donor or acceptor groups from adjacent nucleotides. If a given sequence facilitates the formation of a continuous, properly oriented hydrogen-bond network, proton transfer can proceed with a lower reorganization penalty, which can reduce the activation barrier [27].

DNA dynamics provides a third example. Local micro-motions and transient opening events vary with sequence and control how frequently the target base samples geometries that increase water accessibility and enable the solvent/hydrogen-bond rearrangements required for reaction progress [27].

Experimental observations support the relevance of such sequence effects. CpG sites are well established as mutational hotspots, particularly after methylation of cytosine to 5-mC, consistent with the higher intrinsic propensity of 5-mC to deaminate relative to cytosine in double-stranded DNA [2,28]. Importantly, substitution rates in genomes depend not only on the central CpG site but also on the immediate flanking nucleotides (N−1 and N+1), indicating that nearest-neighbor context modulates the observed frequency of C→T transitions [23].

Spontaneous deamination in DNA is a reaction whose kinetics depend on many factors, including the properties of the substrate itself and the local microenvironment of the helix [2,14]. As a result, even systems that differ only in the composition of nucleotides adjacent to the reaction center may exhibit different activation barriers, as the distribution of electrostatic potential, the organization of hydrogen bonds and stacking interactions, among other things, are altered [25,27]. For this reason, inferring the stabilization of the transition state based on geometry alone is sometimes insufficient, as the presence of a hydrogen bond does not determine the preferential stabilization of the transition state (TS), and even slight changes in the orientation of functional groups and charge distribution can lead to differences in interaction energy that are significant in terms of reaction rate [31].

For this reason, computational methods are used to describe the system to quantitatively assess the stabilization of the substrate and the transition state by the environment. One example of this approach is differential transition-state stabilization (DTSS), a method that treats the effect of the sequence as the differential stabilization of the transition state relative to the substrate [32]. In other words, this technique checks whether a given environment stabilizes the transition state (TS) more strongly than the substrate. If neighboring bases change the stacking, the arrangement of water molecules and the local electrostatic field, they can support the TS of deamination to varying degrees. This difference can be calculated and recorded as a shift in the barrier depending on the sequence context, e.g. in the form of $\Delta\Delta E^{\ddagger}$ values assigned to specific nearest-neighbor systems.

1.4. Aim of the thesis

Based on the key chemical drivers of hydrolytic deamination and the evidence that DNA sequence context modulates reactivity, this thesis aims to quantify how the immediate nucleotide environment affects the activation barrier, and thus the expected rate, of cytosine and 5-methylcytosine deamination. The main goal is to identify local sequence patterns adjacent to the deamination site that promote the reaction, understood here as contexts that lower the barrier by stabilizing the transition state more strongly than the reactant state.

To accomplish this, representative structural models of both the substrate and the transition state will be selected from published literature and used as consistent reference points for subsequent analyses. Using these structures, the differential transition-state stabilization (DTSS) framework will be applied to evaluate how successive neighboring nucleotides (treated as a controlled variable while the central G-C or G-5mC pair is kept fixed) shift the activation barrier. In practice, calculations will be used to verify whether a given nucleotide residue contributes more to binding of the transition state than to the substrate, which directly corresponds to a reduced barrier in the context of a given sequence.

All computations will be carried out for double-stranded DNA in the A and B conformation to maintain structurally realistic conditions for the process under study. Binding energies for the substrate and the transition state will be calculated using the MED model on the selected reference structures, and the resulting values will provide the basis for DTSS determination. This strategy will allow comparison across sequence and identification of adjacent motifs that energetically favor cytosine and 5-mC deamination.

1.5. Computational methods

1.5.1. Quantum-Chemical Background

Quantum chemistry describes the properties of molecules by determining the electronic energy for a given nucleus geometry based on Schrödinger's equation. The interaction energy between two molecules can also be defined as the difference between the total energy of the complex and its components, with these values resulting from the solutions to the relevant Schrödinger equations. In practice, however, for multi-electron systems, this equation cannot be solved analytically. Therefore, approximate methods of known accuracy are used, selected with the aim of a reliable result at an acceptable computational cost [33,34]

1.5.2. Wave Function and Density-Based Methods

In wave function-based approaches, such as Hartree-Fock, the multi-electron state is described directly by an approximated wave function. In practice, this leads to the inclusion of the average field and the omission of a significant part of the electron correlation. One consequence of this is a less accurate description of the energy compared to experimental data, and improving the quality requires the use of methods such as post-HF that take the above correlation into account, which quickly become very computationally expensive as the size of the system increases [35,36]. [35], [36]

Density functional theory (DFT) takes a different perspective, treating electron density as a fundamental quantity, which makes it practical for medium-to-large chemical systems and often provides a favorable balance between accuracy and computational cost. Hohenberg and Kohn's theorem shows that electron density uniquely determines the ground state energy, and the 'correct' density is the one for which the energy is smallest [35].

In practical implementations, such as Kohn-Sham, the many-electron problem is replaced by a set of equations for electrons moving in an effective potential. These equations are solved iteratively until the resulting density stops changing [20].

For DFT, the bottleneck is both the limited accuracy of functional approximations and the cost of selected calculation steps. In particular, many commonly used density functional approximations (DFAs) do not correctly reproduce the long-range electron correlation responsible for dispersion (London forces). Therefore, in non-covalent systems, dispersion corrections are usually applied, although they require adjustment to a given functional and may lead to phenomena such as overbinding. Furthermore, DFA errors intensify in areas of rapidly changing density and at the 'edges' of the distribution, which worsens the description of geometry and binding energy. In practice, systematic deviations are also observed, such as minor errors in hydrogen bond lengths and a strong dependence of relative energies (e.g., tautomer stability) on the choice of functional and basis set, which often necessitates calibration against experimental data or post-HF methods. Additionally, despite favorable scaling, the cost of DFT is increased by numerical issues, including the integration of exchange-correlation contributions on 3D grids and the calculation of Coulombic components, which is usually accelerated by RI-type methods. [36].

1.5.3. Interaction Energy and the Supermolecular Approach

In the context of the influence of the DNA sequence microenvironment, the interaction energy between the reaction center (substrate or transition state) and fragments of the environment is crucial. The most basic approach to determining this energy is the supermolecular approach, in which the interaction energy between two fragments A and B is defined as the difference between the energy of the dimer and the energy of the isolated monomers:

$$\Delta E_{int} = E_{AB} - E_A - E_B$$

where, E_{AB} is the total electronic energy of the combined system in the geometry of the complex, while E_A and E_B are the total electronic energies of the isolated fragments A and B. With this convention, E_{int} quantifies how much stabilization or destabilization arises from the interaction between A and B. A negative value means that the complex is lower in energy than the separated fragments and the interaction is attractive, while a positive value indicates a net repulsion at that geometry [34].

Despite its wide applicability, this method proves to be challenging in practical implementation for weakly interacting systems, because ΔE_{int} is obtained by subtracting large values from each other, and the expected precision of the order of 10^{-5} to 10^{-7} atomic units is numerically demanding [33]. An additional problem with the above approach is the Basis set Superposition Error (BSSE), which must be controlled, mainly through the counterpoise correction classically proposed by Boys and Bernardi [33]. Regardless of these limitations, the supermolecular approach is a key element of the multi-body expansion (MBE) method. It allows the approximate energy of a large system to be determined by summing the energies of its individual components and corrections resulting from interactions between them [34].

1.5.4. Energy Decomposition Analysis and the HVPT Scheme

In research on biological systems, simply determining energy is often insufficient, and it is crucial to understand the mechanisms that actually stabilize the substrate and the transition state. Although the supermolecular approach is widely used, it only provides a total energy value without offering insight into the physical nature of the interactions. The solution to this problem are interaction energy decomposition analysis (EDA) methods, which allow the total energy to be broken down into specific, interpretable components. This allows to accurately assess what is behind the stabilization of the system. Namely, whether it is the dominant electrostatic forces or whether other effects, such as dispersion interactions, are also significant [31,34].

One such approach is the hybrid variational-perturbational procedure (HVPT) [37]. Within the HVPT scheme, the total interaction energy (e.g., at the MP2 level) is broken down into the following components:

$$\Delta E = E^{(1)}_{EL-MTP} + E^{(1)}_{EL-PEN} + E^{(1)}_{EX} + E^{(R)}_{DEL} + E_{CORR}$$

where the term $E^{(1)}_{EL-MTP}$ corresponds to the first-order electrostatic component described through multipole–multipole interactions between the fragments. The subsequent electrostatic term $E^{(1)}_{EL-PEN}$ captures short-range penetration effects, which become important when the electronic charge densities overlap and the pure multipole approximation is no longer sufficient. The quantity $E^{(1)}_{EX}$ denotes the first-order exchange contribution associated with antisymmetrization of the wavefunction (Pauli repulsion). Delocalization effects are collected in $E^{(R)}_{DEL}$, reflecting energy lowering due to polarization and charge-transfer–type relaxation of the fragments in each other's presence. Finally, E_{CORR} represents the correlation component beyond the SCF level, including dispersion and exchange-dispersion terms that are essential for describing van der Waals interactions. This decomposition allows us to build a hierarchy of models, from more complete to increasingly simpler approximations, and consciously select those levels of theory that determine the observed trends in a given class of systems [31].

1.5.5. Differential Transition-State Stabilization

The results obtained using the HVPT scheme are directly applicable in the DTSS (Differential Transition State Stabilization) approach to the quantitative description of the influence of the environment on the reaction barrier. The basis of this approach is the assumption that the catalytic effect of the environment is not determined by how strongly it stabilizes the reacting system in general, but by whether it stabilizes the transition state more than the substrate, since such preferential stabilization directly translates into a lower activation barrier. . In this framework, the catalytic effect of the environment is expressed as:

$$\Delta_{DTSS} = \Delta\mathrm{E}_{TS} - \Delta\mathrm{E}_{S}$$

where Δ_{DTSS} denotes differential transition-state stabilization, ΔE_{TS} is the stabilization energy of the transition state (TS) due to the environment, and ΔE_S is the stabilization energy of the substrate (S) due to the same environment. This quantity measures the preferential stabilization of TS relative to S. Although, formally, this quantity is related to free energies, the simplified form based on interaction energies is justified when closely related systems, such as mutant series or different DNA sequence contexts, are compared, because vibrational and entropic contributions are largely reduced in differential terms. The main catalytic trends therefore remain correlated with changes in electronic interactions and, in many cases, even with the electrostatic contribution itself. [31].

The combination of DTSS with energy decomposition allows us to move from simple observation of stabilization to a full explanation of its physical mechanism. Analysis of individual components allows us to precisely identify which types of interactions are responsible for lowering the barrier and whether this process is driven mainly by electrostatics or requires other effects to be taken into account. As part of the electrostatic interpretation, DTSS also enables the construction of so-called catalytic fields, determined on the basis of differences in electrostatic potential and electric field between the transition state and the substrate. These fields provide a valuable visual clue, illustrating the distribution of charges in the environment that would preferentially stabilize the TS and would most optimal accelerate the reaction [31,38].

1.5.6. The MED Model

The next step in the hierarchy of computational methods, allowing for the practical application of conclusions drawn from energy decomposition, is the Multipole Electrostatic and Dispersion (MED) model. It is a direct synthesis of the theoretical foundations of energy decomposition with the computational efficiency required for the study of complex biological systems. Full quantum calculations, such as the MP2 method used in the HVPT scheme, prove too costly for the analysis of macromolecules or searching extensive conformational libraries due to the scaling of costs of the order of $O(N^5)$. The MED model solves this problem by building on a key observation from decomposition analyses (EDA) that show that electrostatic interactions dominate in many active centres, whereas other short-range contributions, such as exchange, penetration, and delocalization, as well as electron correlation, often partially cancel each other out [39].

As a result, the MED model reduces the description of interactions to two most important long-range components: multipole electrostatic energy $E^{(1)}_{EL,MTP}$ and dispersion energy D_{as}. Accordingly, the interaction energy in the MED model can be obtained as:

$$\Delta \mathrm{E} \approx \Delta \mathrm{E}^{(1)}_{EL,MTP} + \mathrm{D}_{as}$$

where $\Delta\mathrm{E}^{(1)}_{EL,MTP}$ represents the interaction of permanent multipole moments of monomers and is effectively determined using cumulative atomic multipole moments (CAMM), which avoids time-consuming multi-centre integrals [39], [40]. The dispersion component, on the other hand, complements the description of London interactions, which are missing in the standard Hartree-Fock approach and in semi-local DFT functionals. It is calculated using an atom-atom function fitted to the precise dispersion energies obtained from symmetry-adapted perturbation theory (SAPT) calculations using the Tang-Toennies damping function. The latest version of this model, D^{20}_{as}, extends the parameterisation to new elements, including halogens, which is crucial in the design of modern drugs and the analysis of non-covalent interactions [39].

The main advantage of the MED approach is its computational efficiency resulting from scaling of the order of $O(A^2)$, where A is the number of atoms in the system. This makes the method comparable in computational cost to empirical functions, while preserving its non-empirical character. The model is also characterized by relative sensitivity to geometry errors, i.e. in situations where structures exhibit unphysically short intermolecular distances. In such cases more accurate levels of theory often fail due to the artificial dominance of exchange repulsion, while MED correctly predicts stability rankings.

In practice, the MED model can be applied to the process of designing new drugs and proteins, enabling, among other things, the determination of differential intermediate state stabilization (DISS), which is a measure of intermediate stabilization relative to the substrate analogous to DTSS. As a result, we are able to instantly search through thousands of possible amino acid arrangements (rotamers) to identify conformations that maximize the stabilization of the transition or intermediate state over substrate [39].

Despite its high ranking efficiency, the MED model has certain limitations resulting from the physical simplifications adopted. The deliberate omission of short-range components, such as exchange energy and induction, is based on the assumption that they compensate each other, which may not be the case in every molecular system. The model may also show lower accuracy in systems where bonding is strongly controlled by electron delocalization effects. Furthermore, MED only estimates the relative binding free energy, so its reliability in predicting affinity constants depends on whether the entropy and solvation contributions remain at a similar level in a given series of studied systems. The final significant limitation is the parameterization range of the D^{20}_{as} function, which currently does not include selected metals (such as Li, Na or Mg) due to the difficulty in describing them accurately using simplified pair functions [39].

2. Methodology

2.1. Selection of substrate and transition state models

The substrate and transition-state models used in this study are summarized in Tab. 1. These structures were selected from the theoretical work of Labet et al. on the hydrolytic deamination of cytosine and 5-methylcytosine in an aqueous environment that facilitates proton transfer [41]. The selected structures correspond to hydrolytic deamination in a neutral aqueous environment. For cytosine, the substrate and transition state models were taken from pathway B, whereas for 5-methylcytosine, the corresponding models were taken from pathway 5mB. These pathways describe the reaction of the neutral nucleobase with two water molecules, where one water molecule acts as the nucleophile and the second one assists proton transfer during formation of the tetrahedral intermediate [41].

In the selected mechanism, the first reaction step consists of a nucleophilic addition of one water molecule to carbon C4 of cytosine or 5-methylcytosine. This process is assisted by a second water molecule, so the reacting model includes a water dimer. During this step, proton transfers occur within the water-assisted reaction complex, and a tetrahedral intermediate is formed. This elementary step proceeds through the transition state TS4 for cytosine and 5mTS4 for 5-methylcytosine, which represent the rate determining transition states of pathways B/5mB [41].

For cytosine deamination, the substrate model was taken as R2’, and the corresponding transition-state model was TS4. For 5-methylcytosine deamination, the substrate model was taken as 5mR2’, and the corresponding transition state model was 5mTS4, which are shown in Fig. 3 [41]. These structures were used as fixed reacting fragments in the subsequent interaction energy calculations with neighboring nucleobases.

The selected transition states correspond to the rate determining nucleophilic addition step. In aqueous solution, the activation free energy reported for this step is 138.5 kJ mol^{-1} for cytosine and 134.1 kJ mol^{-1} for 5mC [41]. Therefore, TS4 and 5mTS4 were used as transition state models for evaluating which neighboring DNA bases preferentially stabilize the transition state relative to the substrate.

Tab. 1 Selected substrate and transition state models for cytosine and 5-methylcytosine deamination, together with the corresponding activation free energies reported in aqueous solution. All data taken from Ref. [41].

Model	**Pathway**	**pH / environment**	**Number of H_2O molecules**	**Selected substrate**	**Selected transition state**	**Activation free energy ΔG‡ in aqueous solution**
Cytosine	Pathway B	Neutral aqueous environment in pH 7.4 to 7.8	2 H_2O	R2’	TS4	138.5 kJ mol^{-1}
5mC	Pathway 5mB	Neutral aqueous environment	2 H_2O	5mR2’	5mTS4	134.1 kJ mol^{-1}

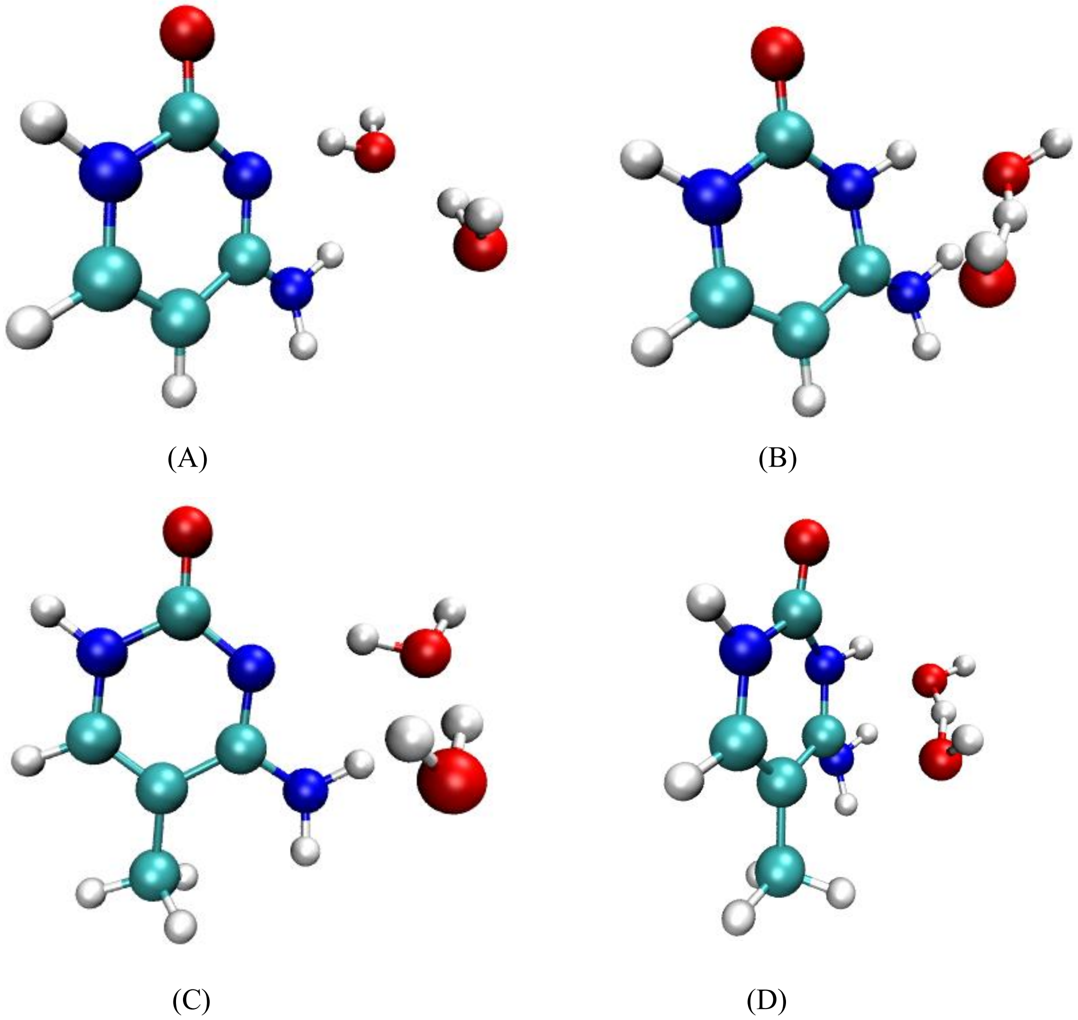

Fig. 3 Substrate and transition state structures selected for the interaction energy calculations: (A) cytosine substrate model R2', (B) cytosine transition state model TS4, (C) 5mC substrate model 5mR2', and (D) 5mC transition state model 5mTS4. All data taken from Ref. [41].

2.2. Preparation of A-DNA and B-DNA model structures

Double stranded DNA models were generated using the X3DNA web server [42]. Two DNA conformations were prepared, corresponding to A-DNA and B-DNA, in order to examine the influence of DNA geometry on the calculated sequence effects.

The use of both forms was justified by the conformational polymorphism of DNA. Although B-DNA is the dominant form under physiological conditions, DNA is not a static structure and may adopt alternative conformations depending on sequence, hydration, methylation and local structural constraints [43,44]. In particular, methylation of cytosine rich CG regions has been reported to promote local transitions toward A-DNA like geometry [43]. A-DNA and B-DNA also differ substantially in their spatial parameters, including base pair inclination, helical pitch and relative arrangement of neighboring bases [43]. Therefore, comparison of both conformations made it possible to evaluate whether the stabilizing effect of neighboring nucleotides depends only on base identity or also on the three dimensional arrangement of the DNA strand.

In both cases, the initial model contained eleven cytosine residues. The central residue, located at position 6, was defined as the reaction site, whereas the five residues on each side were treated as the local sequence environment potentially affecting the deamination process.

2.3. Generation of single nucleotide sequence variants

After generation, the double stranded models were reduced to single strand DNA containing the reacting cytosine and were used in the subsequent stages of the workflow to evaluate the direct interaction between individual neighboring nucleobases and the reacting fragment. This choice was justified by experimental observations showing that the double helix provides protection against cytosine deamination. The half-life of cytosine at 37°C was reported to be about 200 years in single strand DNA, whereas in double strand DNA it is of the order of 30,000 years [4]. In addition, cytosine deamination in single strand DNA has been reported to be 200 to 300 times faster than in double strand DNA, because the reacting base is not protected by the complementary strand [1].

Each neighboring position was mutated separately in order to generate single nucleotide variants of the model DNA fragment. The analyzed positions included residues 1 to 5 and 7 to 11, while residue 6 was kept as the reaction site. For each position, the original cytosine residue was replaced with adenine, thymine or guanine, one mutation at a time. In this way, each generated structure differed from the reference model by only one nucleotide. Representative examples of the generated single nucleotide variants are shown in Fig. 4. The figure presents A-DNA containing the cytosine substrate model at position 6 with four variants of mutations at position 7, and B-DNA containing the 5-methylcytosine substrate model at position 6 with four variants of nucleotides at position 4.

Next, individual mutated nucleotides were isolated from the DNA models for further interaction energy calculations. Only the nucleobase part of each nucleotide was retained, while the remaining part of the nucleotide was removed. Hydrogen atoms were added at the missing valence positions to obtain chemically complete nucleobase fragments.

For each DNA conformation, the complete design initially included ten neighboring positions and four possible bases at each position. Therefore, the planned set consisted of 80 structures for the A-DNA and B-DNA models.

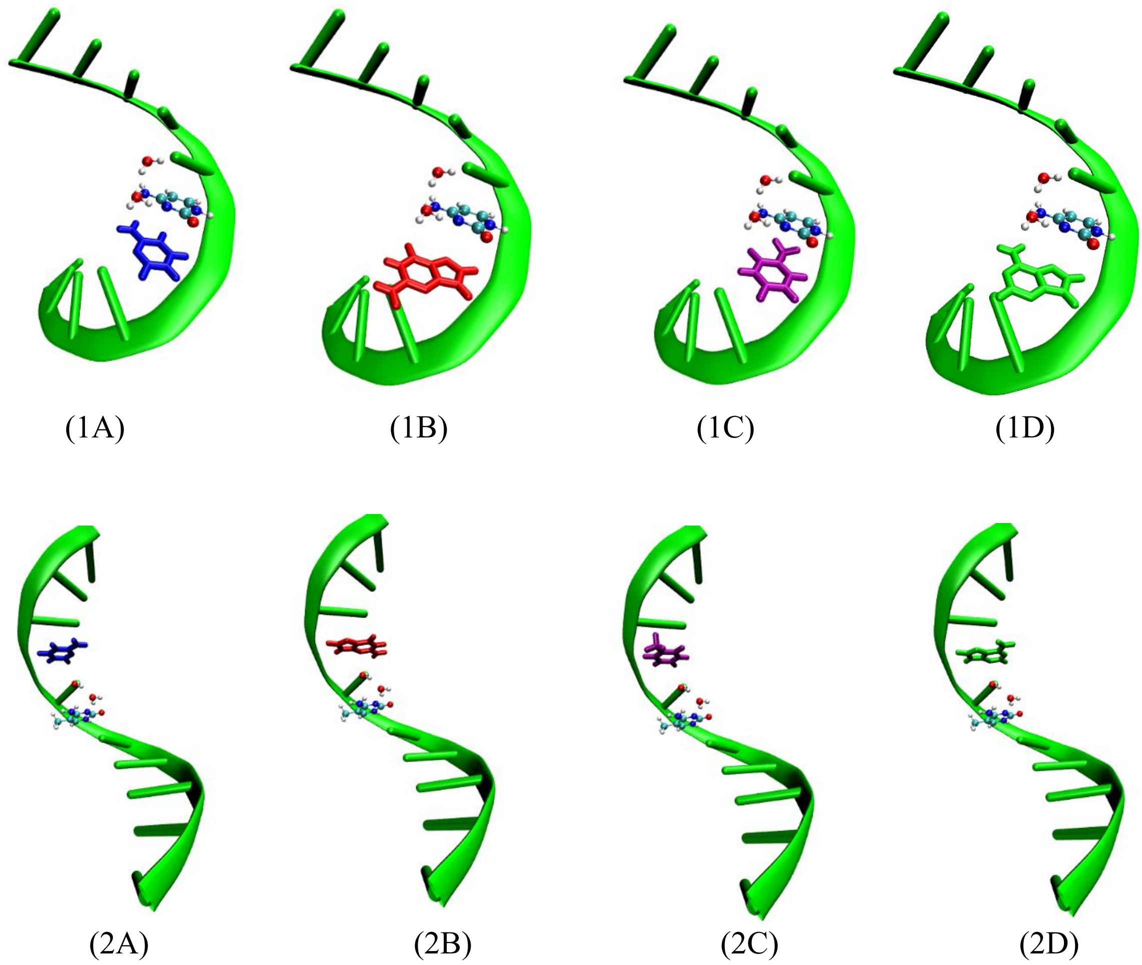

Fig. 4 Representative examples of single nucleotide variants generated for the DNA models. The upper row shows A-DNA structures containing the cytosine substrate model at the reaction site, whereas the lower row shows B-DNA structures containing the 5-methylcytosine substrate model. Panels 1A to 1D present variants of A-DNA, and panels 2A to 2D present variants of B- DNA. Nucleobases are colored according to base identity: cytosine in blue, guanine in red, thymine in purple and adenine in green.

2.4. Alignment of substrate and transition state structures at the reaction site

After preparation of the DNA sequence variants, the central residue at position 6 was replaced by the selected substrate or transition state model acquired from Ref. [41]. The replacement was performed separately for cytosine and 5-methylcytosine deamination models, and separately for the substrate and transition state geometries.

The substrate and transition state structures were superimposed with the position of the central cytosine in the DNA model. The alignment was performed using only the heavy atoms of the pyrimidine ring as reference atoms, since this part of the molecule is common to the substrate and transition state structures. This approach allowed the reacting fragment to be positioned in place of the original cytosine residue, while preserving a comparable orientation within the DNA model.

2.5. MED interaction energy calculations

Interaction energies between each neighboring nucleobase and the reacting fragment were calculated using the MED model [39]. In this approach, the total interaction energy was obtained as the sum of the CAMM multipole electrostatic contribution and the D_{as} dispersion contribution, according to the expression described in the theoretical background section. The CAMM component was calculated using GAMESS (US) version 31 July 2022 [46], whereas the D_{as} component was obtained using dedicated Python scripts [39]. The two contributions were then summed to obtain the final MED interaction energy for each nucleobase fragment pair.

2.6. Differential transition state stabilization analysis

Differential transition state stabilization values were calculated for each neighboring nucleobase by comparing its MED interaction energy with the transition state model and with the corresponding substrate model. This allowed the effect of each neighboring base on the relative stabilization of the transition state to be evaluated.

DTSS values were calculated independently for each base type, each analyzed position, each DNA conformation and each reacting system. This made it possible to compare the effects of different neighboring nucleotides and to identify positions where sequence dependent stabilization was most pronounced.

2.7. Data filtering and visualization

Results obtained for nucleotide position 5 were excluded from the final analysis because this position produced steric hindrance with the substrate and transition state models. In these structures, the water molecules present in the selected substrate and transition state models overlapped with the nucleotide occupying position 5 in the DNA model. This spatial arrangement led to unrealistic contacts between atoms and could distort the calculated interaction energies. These steric effects were treated as artifacts of the model geometry rather than meaningful sequence dependent interactions. Therefore, position 5 was omitted from the final comparison of MED and DTSS values.

For data analysis and visualization, the original nucleotide numbering used in the structural models was converted into a relative numbering system with respect to the central cytosine or 5-methylcytosine (Tab. 2). The reactive C/5mC was defined as position 0. Nucleotides located on the 5′ side of the central base were assigned negative values, whereas nucleotides located on the 3′ side were assigned positive values. Thus, positions 1 to 5 in the original model corresponded to relative positions -5 to -1, while positions 7 to 11 corresponded to +1 to +5.

Tab. 2 Conversion of nucleotide numbering in the starting DNA model to positions relative to the central C/5mC.

Number in the DNA model	Position relative to central C/5mC
1	−5
2	−4
3	−3
4	−2
5	−1
6	0, central C/5mC
7	+1
8	+2
9	+3
10	+4
11	+5

3. Results and discussion

In the following discussion, nucleotide positions are reported relative to the central cytosine and 5-methylcytosine, which was defined as position 0. Therefore, negative values indicate residues located before the reactive base, whereas positive values indicate residues located after it in the DNA sequence, assuming 5' to 3' direction of DNA strand. The nucleotide at position -1 was excluded from the analysis due to steric clashes with transition state and substrate model observed in the corresponding complexes.

3.1. MED interaction energies

The MED interaction energies are presented as heatmaps in Fig. 5 and as interaction energy profiles in Fig. 6. The heatmaps provide a detailed comparison of MED values for individual neighboring nucleotides and positions. The profiles show the overall positional trends more clearly. The numerical MED values and their D_{as} and CAMM components are provided in Appendix, Tables 3-6.

In all analyzed systems, as shown in Fig.5 and Fig. 6, the strongest interactions were observed mainly for position +1, directly adjacent to the reaction center. This trend was present for both cytosine and 5-methylcytosine models, as well as for A-DNA and B-DNA structures. Among the neighboring nucleotides, thymine at position +1 usually showed the most negative MED values, indicating the strongest attractive interaction with the reaction model.

For example, in the A-DNA substrate model, (+1)T showed MED values of -9.8 kcal/mol for the cytosine and -10.6 kcal/mol for the 5-methylcytosine. These negative values indicate that (+1)T forms the most favorable interaction with the reaction model located at the central position, suggesting that the nucleotide directly adjacent to the deamination site has the largest contribution to the MED overall binding energy.

In contrast, positions located farther from the reaction center, especially positions -5, -4, +3, +4 and +5, showed MED values close to zero, indicating only weak interactions with the reaction model. This pattern is particularly clear in the interaction energy profiles shown in Fig. 6, where the curves become almost flat at positions distant from the central C/5mC reactants. Moderate effects were observed for positions -2, +1 and +2, which are located closer to the deamination site. Therefore, the MED results suggest that the interaction energy is mainly affected by the local nucleotide environment around the central C/5mC. This agrees with the nearest-neighbor description of DNA thermodynamics, according to which the energetic properties of DNA depend strongly on neighboring base pairs and local stacking interactions [47].

In this work, A-DNA and B-DNA were used as predefined structural frameworks to examine how the identity of neighboring nucleobases affects their interaction with the reaction model in different DNA geometries. Therefore, the observed differences between A-DNA and B-DNA should not be interpreted as sequence-induced changes in DNA structure, but rather as the effect of placing the same set of nucleobases in two distinct idealized conformations. A-DNA models showed a broader range of MED values than B-DNA models. This was especially visible at position -2, where guanine produced positive MED values in A-DNA substrate models, reaching +3.3 kcal/mol for cytosine and +3.2 kcal/mol for 5-methylcytosine. Such positive values indicate unfavorable interactions. In B-DNA, the corresponding values were much smaller, suggesting a more moderate interaction in this DNA form. Thus, the comparison between A-DNA and B-DNA indicates that the calculated interaction energies are sensitive not only to nucleobase identity, but also to the geometric framework in which the interaction is evaluated.

The comparison between cytosine and 5-methylcytosine showed that methylation does not substantially change the general distribution of MED values, since position +1 remains the dominant interaction site in both cases. However, selected interactions, especially those involving thymine at relative position +1, denoted as (+1)T, were slightly more negative in the 5mC models. This suggests that the presence of the methyl group may locally affect the interaction strength, although the main sequence-dependent pattern remains preserved.

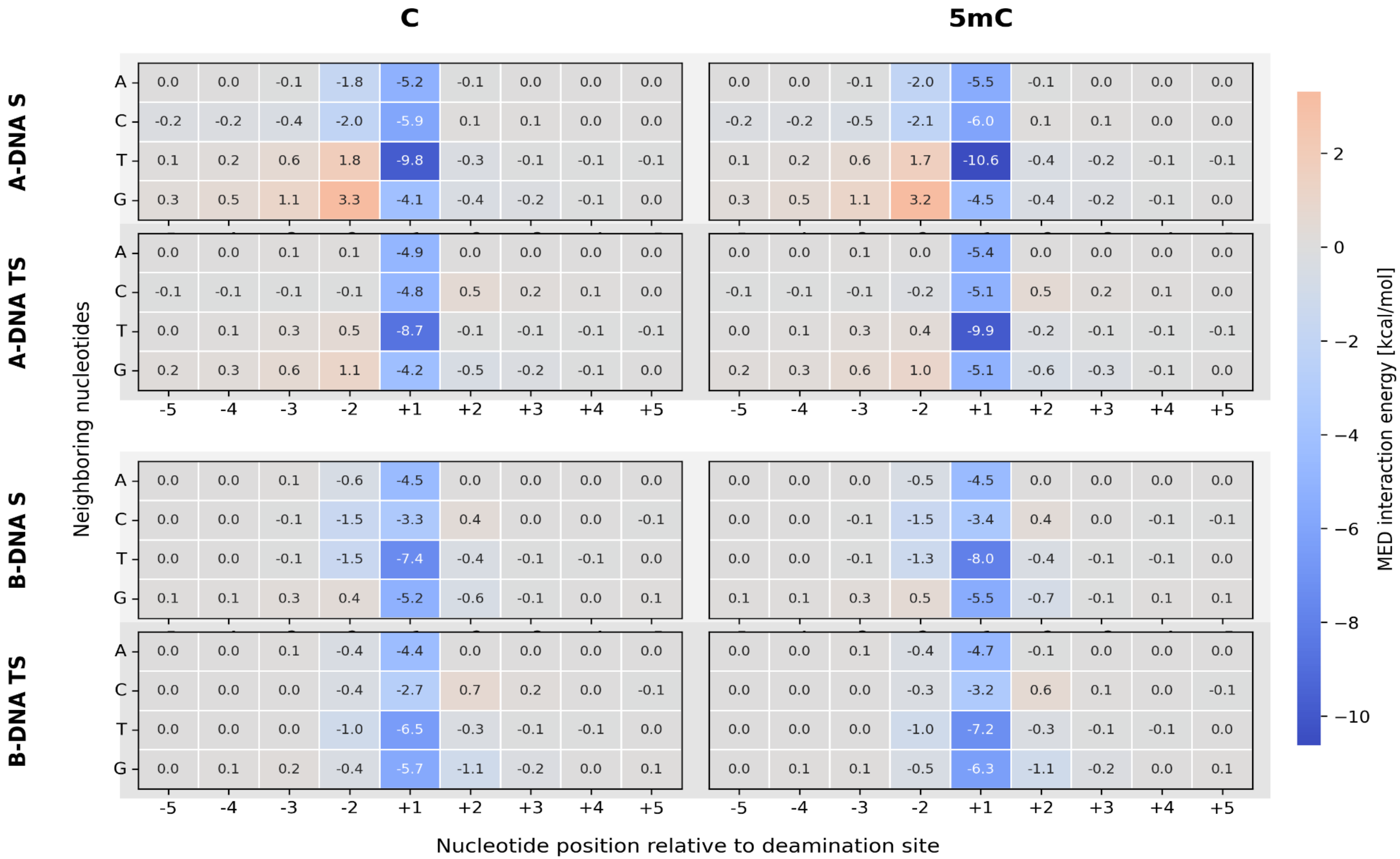

Fig. 5 Heatmaps of MED interaction energies for substrate (S) and transition state (TS) models of C and 5mC deamination in A and B-DNA models.

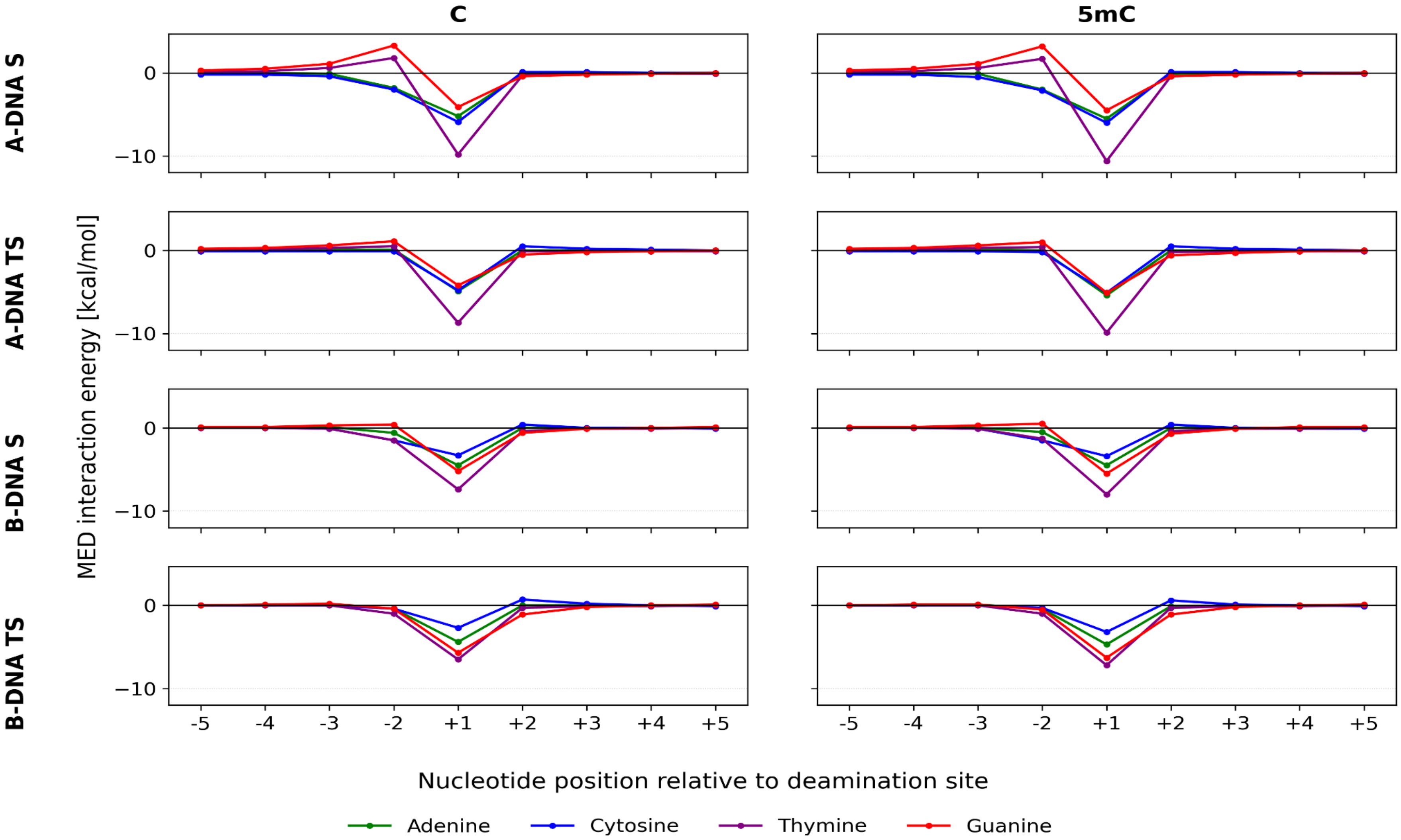

Fig. 6 MED interaction energy profiles for substrate (S) and transition state (TS) models of C and 5mC deamination in A-DNA and B-DNA. The lines represent distinct nucleotide type at positions numbered relative to the deamination site.

3.2. Physical nature of interactions

The physical nature of the reactant binding was further analyzed by comparing the D_{as} and CAMM components for the most relevant nucleotide positions, namely -2, +1 and +2 (Fig. 7). Nucleotides occupying more distant positions relative to deamination site were not included in this analysis due to much weaker reactant binding. Since the analysis of transition state binding yields similar conditions as drawn in what follows for substrate binding, only the latter is discussed here. As shown in Fig. 7 the strongest attractive MED interactions observed at position +1 are mainly driven by the D_{as} component. In case of the T nucleotide occupying +1 position in the A-DNA substrate model, the D_{as} contribution was highly negative, whereas the CAMM contribution was much smaller and. As a result, the favorable MED value for this nucleotide was dominated by the dispersion term.

A similar tendency was observed for both cytosine and 5-methylcytosine models, as well as for A-DNA and B-DNA structures. In most cases at position +1, the D_{as} component was negative and had a larger absolute value than CAMM, indicating that dispersion interactions were the main source of stabilization. This can be explained by the close contact between the reaction model and neighboring nucleotides. DNA bases contain extended aromatic systems and are relatively polarizable, which makes dispersion interactions important, especially when the fragments are located close to each other. Since the D_{as} term represents the dispersion contribution, it is generally attractive and can dominate the total MED interaction energy when the geometry allows efficient short range contact between the reaction model and the neighboring base.

The CAMM component reflects the multipole electrostatic contribution and is therefore more sensitive to the relative orientation of the interacting fragments and to the local distribution of electrostatic potential. For this reason, CAMM may either stabilize or destabilize the interaction. Negative CAMM values indicate favorable electrostatic complementarity between the reaction model and the neighboring nucleotide, for example when electron rich regions are oriented toward electron deficient regions. Such stabilizing CAMM contributions were observed, for (+1)G in B-DNA models, where the CAMM values were negative and therefore strengthened the attractive character of the total MED interaction (Fig. 7). In contrast, positive CAMM values suggest an unfavorable electrostatic arrangement, where the multipole interactions partially compensate or even overcome the attractive D_{as} contribution. This effect was especially visible at position -2, where the interactions showed a stronger dependence on the nucleotide type. For guanine and thymine in A-DNA models, the CAMM component was positive and large enough to overcome the attractive D_{as} contribution, leading to positive MED values (Fig. 7). In particular, (-2)G in A-DNA models showed strongly positive CAMM values, which indicates that the unfavorable interaction observed for this system originates mainly from the multipole electrostatic term. For adenine and cytosine at the same position, the CAMM contribution was smaller or negative, resulting in more favorable total MED values. Therefore, position -2 appears to be more sensitive to the electrostatic arrangement of the neighboring base.

At position +2, both D_{as} and CAMM contributions were generally small, which explains why the total MED values were close to zero for most nucleotide types. This confirms that nucleotides located farther from the reaction center contribute much less to the reactant binding. Overall, the analysis of D_{as} and CAMM components of MED interaction energy indicates that the strongest reactant stabilization near the reaction center is mainly dispersion driven, whereas unfavorable or nucleotide dependent effects, especially at position -2, are controlled to a larger extent by the multipole electrostatic contribution.

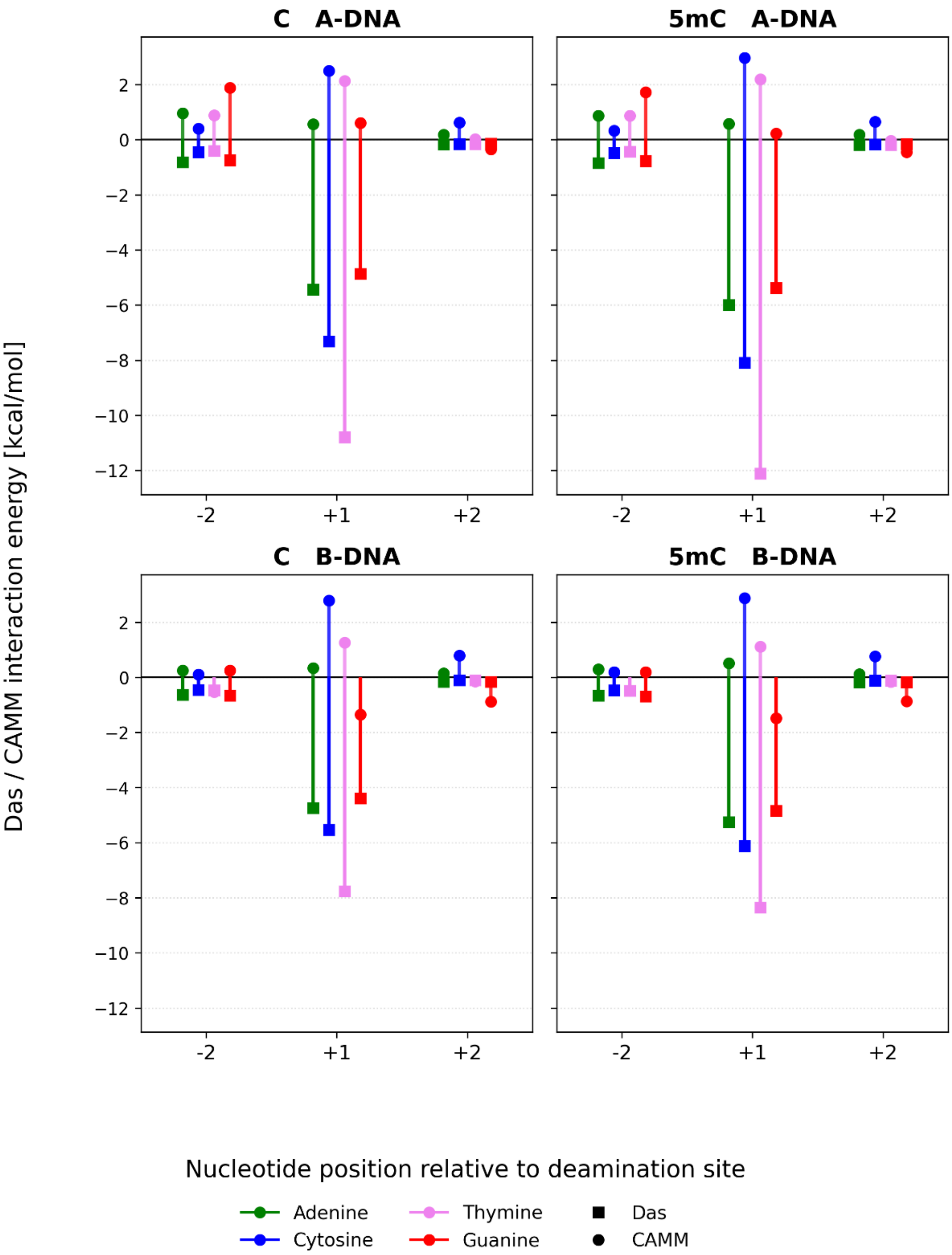

Fig. 7 Comparison of D_{as} (dispersion) and CAMM electrostatic multipole contributions to MED interaction energies for substrate models of cytosine and 5-methylcytosine deamination in A-DNA and B-DNA at selected, most influential neighboring positions around the central cytosine..

3.3. Differential transition state stabilization analysis

3.3.1. Effect of nucleotide position and identity

The DTSS values calculated for cytosine and 5-methylcytosine models in A-DNA and B-DNA are summarized in Fig. 8 . The numerical values are provided in Appendix, Table 7. As shown in Fig. 8, the DTSS analysis revealed that the largest differential stabilization effects are associated mainly with positions -2, +1 and +2 relative to the reaction site. In contrast, distant positions, especially -5, -4, +4 and +5, have values close to zero. Thus, preferential stabilization of the transition state or substrate is mainly controlled by nucleotides located close to the deamination site.

The local character of the DTSS effect agrees with previous analyses of sequence dependent mutagenesis, which described the influence of adjacent nucleotides on single base pair substitutions in human genes [23]. The effect of the surrounding DNA sequence has been reported to be locally limited and to extend no further than two base pairs from the mutation site. A similar concept is used in contemporary mutational signature analysis, where single base substitutions are classified together with the immediately adjacent 5′ and 3′ nucleotides [48]. In this context, the DTSS pattern supports the idea that mainly nucleotides close to the reacting cytosine may contribute to sequence dependent deamination.

Among the analyzed cases, guanine at position -2 showed the most negative DTSS values. This effect was observed for both cytosine and 5-methylcytosine models and in both DNA forms. The strongest negative values were found in A-DNA, reaching -2.2 kcal/mol for both cytosine and 5-methylcytosine deamination. In B-DNA, the effect was weaker, with DTSS values of -0.8 kcal/mol for cytosine and -1.0 kcal/mol for 5-methylcytosine. These values correspond to preferential stabilization of the transition state and suggest that (-2)G may represent a favorable local sequence context for deamination in the analyzed models.

In contrast, cytosine at position -2, yielded positive DTSS values in all systems. The values reached 1.9 kcal/mol in A-DNA both cytosine and 5-methylcytosine, and 1.2 kcal/mol in B-DNA for deamination. This means that (-2)C stabilized the substrate more than the transition state and may therefore represent a less favorable sequence context for deamination. Thymine at the same relative position, denoted as (-2)T, showed a stronger dependence on DNA geometry. In A-DNA, (-2)T stabilized the transition state, with DTSS values of -1.3 kcal/mol for both cytosine and 5-methylcytosine deamination. In B-DNA, however, this effect was not observed, and the corresponding values were positive, reaching 0.5 kcal/mol for cytosine and 0.4 kcal/mol for 5-methylcytosine.The effect of a given nucleotide therefore depends not only on its chemical type, but also on its location relative to the reaction site resulting from the geometry of the DNA helix.

Overall, DTSS-based sequence preferences are summarized schematically in Fig. 9. This representation highlights nucleotides with favorable DTSS effects above the baseline and nucleotides with unfavorable effects below the baseline, making the contrast between favorable and unfavorable local contexts. As emphasized in Fig. 9, the most pronounced favorable effect is associated with guanine at position -2, especially in A-DNA, whereas cytosine and adenine at the same position tend to show unfavorable DTSS effects resulting in decreased deamination rate. The schematic representation in Fig. 9 also shows that the sequence preferences are more pronounced in A-DNA than in B-DNA, where the letters are smaller indicating a weaker impact on the reaction rate.

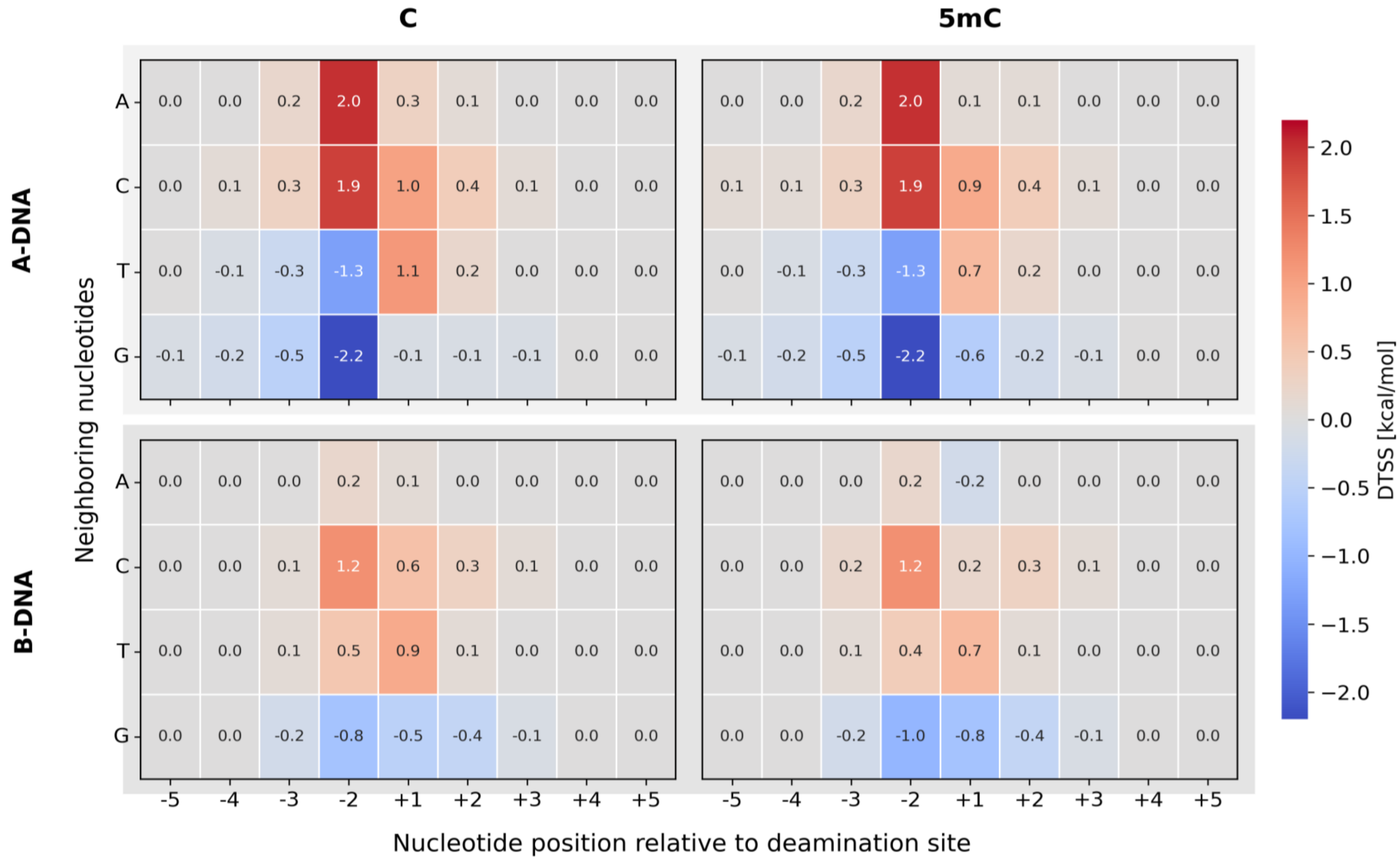

Fig. 8 Heatmaps of DTSS values for cytosine and 5-methylcytosine deamination models in A-DNA and B-DNA.

This position dependent behavior is consistent with the idea that the local sequence context can influence both DNA structure and mutation related processes. Donigan and Sweasy summarized evidence that the bases directly neighboring a mutated site can affect base substitution mutagenesis and base excision repair, including the repair of T:G mismatches formed after 5mC deamination [49]. In the present DTSS results, different neighboring nucleotides also show different effects on preferential transition state stabilization. This suggests that nucleotide identity and position may contribute to sequence dependent deamination by affecting site activation energy barrier.

3.3.2 Differences between C and 5mC deamination models

The comparison between cytosine and 5-methylcytosine models showed that methylation does not completely change the overall DTSS pattern (Fig. 8 and 9). In both models, (-2)G yields negative DTSS values, whereas (-2)C gives positive values. Methylation therefore modifies selected interactions to a minor extent, while the general direction of the DTSS effect remains largely governed by the surrounding sequence context.

Interestingly, some local differences between C and 5mC were observed. Mainly adenine at position (+1)A showed positive DTSS values in the cytosine models, reaching approximately 0.3 kcal/mol in A-DNA and 0.10 kcal/mol in B-DNA. In the 5mC models, this effect became weaker in A-DNA, approximately 0.11 kcal/mol, and changed to a slightly negative value in B-DNA -0.2 kcal/mol. This suggests that methylation can locally modify the balance between substrate and transition state stabilization, although it does not completely reorganize the DTSS sequence-dependent pattern. It should be noted that no definite conclusions can be drawn from energy differences as small as discussed in this paragraph, as it exceeds method accuracy.

However, selected DTSS values become slightly more negative in the 5mC models, especially for (-2)G and (+1)G in B-DNA. This indicates that the methyl group can locally modulate transition state stabilization. Such an effect is chemically reasonable, because cytosine methylation is known to modify local DNA structure and dynamics. Molecular dynamics studies showed that methylation and hydroxymethylation of cytosine can change the dynamical landscape of DNA and affect local twist, roll and tilt parameters near the modified sites [44]. The methyl group at C5 also changes the hydrophobicity of the major groove and modifies local hydration around the methylated base, which may influence local interactions with the reaction model [44]. More recent studies also indicate that cytosine modifications can increase DNA stiffness, mainly by restricting roll flexibility, and that these effects depend on the local sequence context [45].

The biological importance of this comparison is related to the fact that deamination of 5mC produces thymine and may therefore lead to C to T transition mutations. This is different from cytosine deamination, which produces uracil and is usually more efficiently repaired. Classical experimental studies showed that 5mC in double stranded DNA deaminates faster than unmodified cytosine and that methylated cytosines are important mutational targets [2,4]. Methylated CpG sites are also described as mutation hot spots in vertebrate genomes, because deamination of 5mC can convert CpG sites into TpG or CpA products [10,50]. This agrees with modern cancer genomics, where C to T substitutions in NpCpG contexts are associated with age related mutational processes attributed to spontaneous 5mC deamination [48].

For this reason, even small differences in DTSS between cytosine and 5mC models may be relevant when discussing local sequence contexts that could favor 5mC deamination. Nevertheless, these results should be interpreted as a local stabilization tendency rather than as a direct prediction of reaction rates. DTSS compares the interaction energy of a neighboring nucleotide with the transition state and with the substrate, so it identifies whether the local environment preferentially stabilizes the transition state relative to the substrate [31,32]. It does not include all factors that control the full reaction rate, such as conformational sampling, entropy, solvation and DNA repair efficiency.

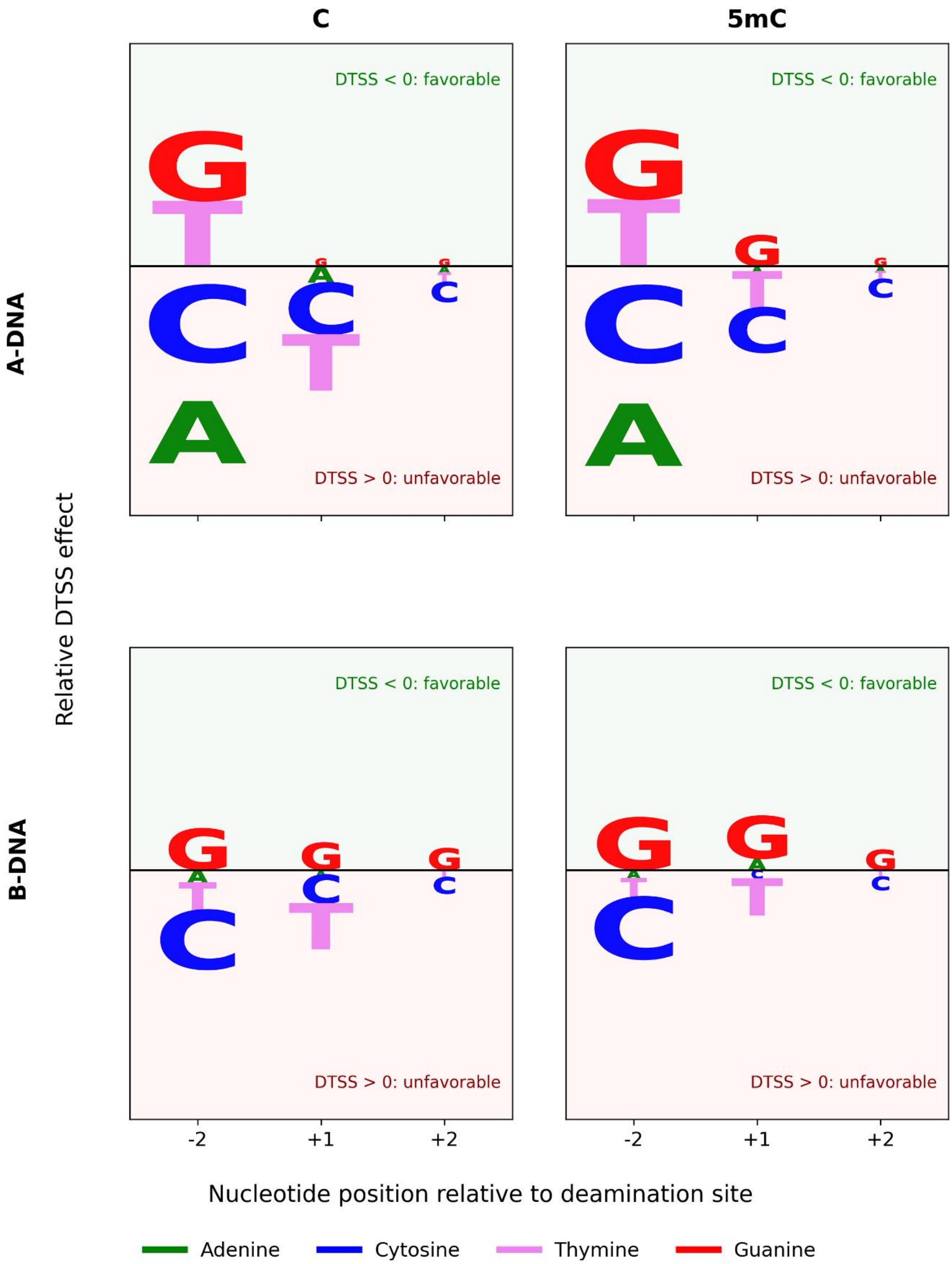

Fig. 9 The catalytic impact of DNA sequence representation patterns around the reacting cytosine or 5-methylcytosine. Letters above the baseline indicate nucleotides with negative DTSS values, corresponding to preferential transition state stabilization and a favorable effect on deamination. Letters below the baseline indicate nucleotides with positive DTSS values, corresponding to preferential substrate stabilization effects, and lowering of deamination rate. The size of letters is proportional to the absolute DTSS values

3.3.3 Differences between A-DNA and B-DNA models

The comparison between A-DNA and B-DNA showed that A-DNA gives larger DTSS amplitudes (Fig. 8). This means that the A-DNA geometry enhances the difference between substrate and transition state stabilization. The strongest example is (-2)G, which reaches more negative values in A-DNA than in B-DNA for both C and 5mC models. Positive DTSS values, such as those observed for (-2)C, are also more pronounced in A-DNA.

A similar tendency was observed for adenine at position -2. In A-DNA, (-2)A gave high positive DTSS values, reaching approximately 2.0 kcal/mol for both cytosine and 5mC models (Fig. 8). In B-DNA, the corresponding values were much smaller. This means that (-2)A preferentially stabilizes the substrate, especially in the A-DNA geometry.

In contrast, B-DNA shows a more moderate DTSS profile. The same general trends are still visible, especially negative values for (-2)G and positive values for (-2)C, but the absolute values are smaller. Interestingly, the tendency toward larger DTSS amplitudes in A-DNA is also visible at position -3. In A-DNA, guanine at position -3 gives clearly negative DTSS values, whereas in B-DNA the values at this position are closer to zero. The helical form therefore affects how strongly neighboring nucleotides differentiate between the substrate and transition state models.

The favorable local sequence context identified from the DTSS analysis is shown in Fig. 10. For the four analyzed positions, namely -3, -2, +1 and +2, the most favorable nucleotide was guanine in all four systems. Thus, the sequence predicted to favor deamination was identical for C and 5mC in both A-DNA and B-DNA and can be written as (-3)G, (-2)G, (+1)G and (+2)G. Although the identity of the favorable nucleotides was conserved, the magnitude of the stabilizing effect differed between the systems. The total DTSS value, calculated as the sum of DTSS values for these four guanine residues, was -2.90 kcal/mol for C in A-DNA, -3.46 kcal/mol for 5mC in A-DNA, -1.97 kcal/mol for C in B-DNA and -2.47 kcal/mol for 5mC in B-DNA (Fig. 10). This indicates that the same local sequence context is predicted to stabilize the transition state in all four systems, but the effect is stronger in A-DNA than in B-DNA and stronger for 5mC than for cytosine within the same DNA conformation.

The opposite, unfavorable sequence context can be defined as the combination of nucleotides giving the largest positive DTSS values, and therefore the strongest preferential stabilization of the substrate. In this case, the pattern was less strictly conserved than for the favorable sequence. Cytosine at position -3 and cytosine at position +2 gave the largest positive contributions in all four systems. However, the most unfavorable nucleotide at position -2 differed between A-DNA and B-DNA, with adenine being dominant in A-DNA and cytosine in B-DNA. At position +1, thymine was the most unfavorable nucleotide in most systems, whereas cytosine gave the largest positive DTSS value for 5mC in A-DNA. Thus, the favorable sequence pattern was more robust across the four systems, while the unfavorable pattern showed stronger dependence on the DNA conformation and on the identity of the reaction site.

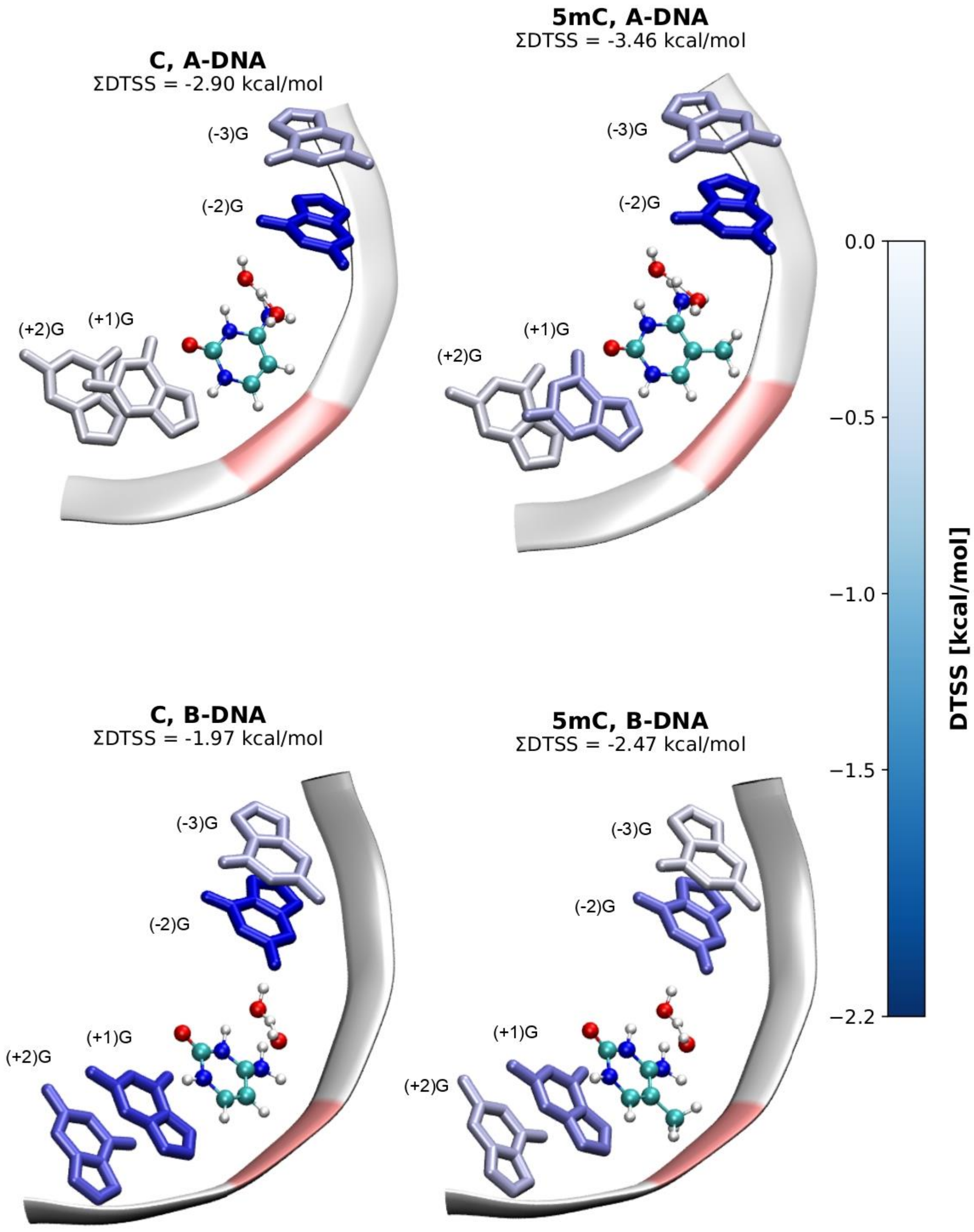

Fig. 10 DTSS based visualization of favorable local sequence contexts for C and 5mC deamination in A-DNA and B-DNA models.

4. Conlcusions

The aim of this work was to evaluate how the local nucleotide sequence affects the interaction pattern and differential transition state stabilization in cytosine and 5-methylcytosine deamination models incorporated into single strand DNA in A-DNA and B-DNA conformations. The results show that the influence of neighboring nucleotides is highly local. Interaction energy analysis indicates that nucleotides located close to the deamination site have the largest effect, whereas more distant positions contribute only weakly. This agrees with the general view that DNA properties are strongly affected by local sequence context and nearest neighbor interactions [47].

According to MED binding energy analysis, the strongest interactions occur mainly at position +1, directly adjacent to the reaction center. In particular, thymine at position +1, denoted as (+1)T, features the most negative MED values in all complexes. The analysis of the physical nature of interactions further showed that these strongest attractive interactions were mainly dominated by the D_{as} component of MED, indicating an important role of dispersion interactions near the reaction center. In contrast, the CAMM multipole electrostatic energy was more sensitive to nucleotide identity and spatial orientation; positive CAMM values weakened or overcame the attractive D_{as} contribution, whereas negative CAMM values strengthened the total interaction.

The DTSS results suggest that guanine at position -1 is the most favorable local context for preferential transition state stabilization in the analyzed models. This effect was observed for both cytosine and 5-methylcytosine and in both DNA forms. In contrast, cytosine at position -1 preferentially stabilized the substrate, seemingly decreasing the reaction rate. Overall, the obtained results support the idea that the effect of a neighboring nucleotide depends not only on the type of base, but also on its position relative to the reaction center. Such a local sequence dependence is consistent with studies showing that neighboring nucleotides influence mutation patterns and that sequence context is commonly considered in mutational signature analysis.

The comparison between cytosine and 5-methylcytosine deamination showed that methylation does not completely reorganize the DTSS pattern. The same general trends were preserved in both models, but selected effects became slightly stronger in the 5mC systems. This suggests that the methyl group can locally modify transition state stabilization without changing the overall sequence dependent behavior. This observation is important because 5mC deamination produces thymine and can therefore lead to C to T transition mutations, especially in methylated CpG contexts, which are known mutation hot spots [2,4,10,50].

The comparison between A-DNA and B-DNA demonstrated that the helical form also affects DTSS values. A-DNA generally produced larger amplitudes, whereas B-DNA showed a more moderate profile. Therefore, the same neighboring nucleotide can influence the substrate and transition state differently depending on the spatial arrangement imposed by the DNA conformation. This indicates that sequence effects should be interpreted together with local DNA geometry.

Overall, the results indicate that local sequence context can modulate cytosine and 5-methylcytosine deamination by changing the relative stabilization of the substrate and transition state. Together, the MED and DTSS analyses show that the local nucleotide environment, methylation status and DNA helical form jointly shape the energetic profile around the deamination site. These findings support the conclusion that sequence dependent deamination cannot be explained by nucleotide identity alone, but requires consideration of the spatial and energetic organization of the surrounding DNA fragment.

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6. Appendix

Tab. 3 Interaction energy components calculated for cytosine in A-DNA. The left part shows values for the substrate model (S), while the right part shows values for the transition state model (TS).

Nucleotide position	Das [kcal/mol]	CAMM [kcal/mol]	MED [kcal/mol]	Das [kcal/mol]	CAMM [kcal/mol]	MED [kcal/mol]
(-5)A	-0.004	0.008	0.004	-0.003	0.007	0.004
(-5)T	-0.003	0.069	0.066	-0.003	0.045	0.042
(-5)C	-0.003	-0.152	-0.155	-0.003	-0.107	-0.11
(-5)G	-0.003	0.289	0.286	-0.003	0.202	0.199
(-4)A	-0.012	0.012	0	-0.010	0.038	0.028
(-4)T	-0.009	0.202	0.193	-0.008	0.128	0.12
(-4)C	-0.009	-0.216	-0.225	-0.008	-0.102	-0.11
(-4)G	-0.011	0.517	0.506	-0.009	0.315	0.306
(-3)A	-0.059	0.007	-0.052	-0.047	0.151	0.104
(-3)T	-0.041	0.600	0.559	-0.035	0.332	0.297
(-3)C	-0.044	-0.372	-0.416	-0.036	-0.076	-0.112
(-3)G	-0.050	1.111	1.061	-0.042	0.643	0.601
(-2)A	-1.482	-0.340	-1.822	-0.819	0.967	0.148
(-2)T	-0.451	2.223	1.772	-0.404	0.891	0.487
(-2)C	-0.619	-1.342	-1.961	-0.448	0.397	-0.051
(-2)G	-1.076	4.382	3.306	-0.742	1.890	1.148
(0)C	0.000	0.000	0.000	0.000	0.000	0.000
(+1)A	-5.279	0.081	-5.198	-5.439	0.559	-4.88
(+1)T	-10.341	0.575	-9.766	-10.798	2.131	-8.667
(+1)C	-6.763	0.904	-5.859	-7.319	2.495	-4.824
(+1)G	-4.859	0.755	-4.104	-4.859	0.609	-4.25
(+2)A	-0.165	0.058	-0.107	-0.164	0.188	0.024
(+2)T	-0.161	-0.159	-0.32	-0.161	0.016	-0.145
(+2)C	-0.148	0.219	0.071	-0.149	0.622	0.473
(+2)G	-0.150	-0.212	-0.362	-0.145	-0.352	-0.497
(+3)A	-0.021	0.030	0.009	-0.021	0.058	0.037
(+3)T	-0.019	-0.117	-0.136	-0.019	-0.102	-0.121
(+3)C	-0.018	0.102	0.084	-0.018	0.205	0.187
(+3)G	-0.019	-0.168	-0.187	-0.019	-0.223	-0.242
(+4)A	-0.005	0.003	-0.002	-0.006	0.014	0.008
(+4)T	-0.004	-0.081	-0.085	-0.004	-0.088	-0.092
(+4)C	-0.004	0.031	0.027	-0.004	0.067	0.063
(+4)G	-0.005	-0.086	-0.091	-0.005	-0.119	-0.124
(+5)A	-0.002	-0.017	-0.019	-0.002	-0.012	-0.014
(+5)T	-0.002	-0.063	-0.065	-0.002	-0.071	-0.073
(+5)C	-0.002	-0.022	-0.024	-0.002	-0.006	-0.008
(+5)G	-0.002	-0.016	-0.018	-0.002	-0.038	-0.04

Tab. 4 Interaction energy components calculated for cytosine in B-DNA. The left part shows values for the substrate model (S), while the right part shows values for the transition state model (TS).

Nucleotide position	Das [kcal/mol]	CAMM [kcal/mol]	MED [kcal/mol]	Das [kcal/mol]	CAMM [kcal/mol]	MED [kcal/mol]
(-5)A	-0.002	0.014	0.012	-0.002	0.015	0.013
(-5)T	-0.001	-0.000	-0.001	-0.001	0.002	0.001
(-5)C	-0.001	0.006	0.005	-0.001	0.006	0.005
(-5)G	-0.002	0.054	0.052	-0.002	0.043	0.041
(-4)A	-0.007	0.034	0.027	-0.006	0.037	0.031
(-4)T	-0.005	0.002	-0.003	-0.005	0.023	0.018
(-4)C	-0.005	0.006	0.001	-0.004	0.023	0.019
(-4)G	-0.008	0.136	0.128	-0.007	0.098	0.091
(-3)A	-0.047	0.098	0.051	-0.036	0.087	0.051
(-3)T	-0.036	-0.077	-0.113	-0.027	0.008	-0.019
(-3)C	-0.034	-0.063	-0.097	-0.026	0.073	0.047
(-3)G	-0.049	0.370	0.321	-0.039	0.189	0.15
(-2)A	-1.327	0.698	-0.629	-0.635	0.255	-0.38
(-2)T	-1.009	-0.521	-1.53	-0.468	-0.534	-1.002
(-2)C	-0.981	-0.558	-1.539	-0.457	0.107	-0.35
(-2)G	-1.329	1.741	0.412	-0.658	0.248	-0.41
(0)C	0.000	0.000	0.000	0.000	0.000	0.000
(+1)A	-4.692	0.195	-4.497	-4.745	0.341	-4.404
(+1)T	-7.718	0.300	-7.418	-7.772	1.254	-6.518
(+1)C	-5.482	2.165	-3.317	-5.532	2.784	-2.748
(+1)G	-4.392	-0.810	-5.202	-4.388	-1.354	-5.742
(+2)A	-0.163	0.115	-0.048	-0.161	0.141	-0.02
(+2)T	-0.108	-0.284	-0.392	-0.106	-0.150	-0.256
(+2)C	-0.107	0.463	0.356	-0.106	0.796	0.69
(+2)G	-0.169	-0.450	-0.619	-0.164	-0.889	-1.053
(+3)A	-0.018	0.009	-0.009	-0.017	0.017	0
(+3)T	-0.012	-0.111	-0.123	-0.012	-0.105	-0.117
(+3)C	-0.012	0.049	0.037	-0.011	0.174	0.163
(+3)G	-0.019	-0.055	-0.074	-0.019	-0.204	-0.223
(+4)A	-0.004	-0.017	-0.021	-0.004	-0.022	-0.026
(+4)T	-0.003	-0.051	-0.054	-0.003	-0.067	-0.07
(+4)C	-0.003	-0.046	-0.049	-0.002	-0.016	-0.018
(+4)G	-0.004	0.052	0.048	-0.004	0.007	0.003
(+5)A	-0.001	-0.019	-0.02	-0.001	-0.025	-0.026
(+5)T	-0.001	-0.029	-0.03	-0.001	-0.044	-0.045
(+5)C	-0.001	-0.063	-0.064	-0.001	-0.066	-0.067
(+5)G	-0.001	0.064	0.063	-0.001	0.058	0.057

Tab. 5 Interaction energy components calculated for 5-methylcytosine in A-DNA. The left part shows values for the substrate model (S), while the right part shows values for the transition state model (TS).

Nucleotide position	Das [kcal/mol]	CAMM [kcal/mol]	MED [kcal/mol]	Das [kcal/mol]	CAMM [kcal/mol]	MED [kcal/mol]
(-5)A	-0.004	0.006	0.002	-0.003	0.006	0.003
(-5)T	-0.004	0.064	0.06	-0.003	0.045	0.042
(-5)C	-0.003	-0.161	-0.164	-0.003	-0.106	-0.109
(-5)G	-0.004	0.294	0.29	-0.003	0.198	0.195
(-4)A	-0.013	0.006	-0.007	-0.011	0.033	0.022
(-4)T	-0.010	0.198	0.188	-0.008	0.125	0.117
(-4)C	-0.010	-0.233	-0.243	-0.008	-0.103	-0.111
(-4)G	-0.012	0.527	0.515	-0.010	0.305	0.295
(-3)A	-0.064	-0.013	-0.077	-0.050	0.135	0.085
(-3)T	-0.045	0.602	0.557	-0.038	0.324	0.286
(-3)C	-0.048	-0.403	-0.451	-0.038	-0.082	-0.12
(-3)G	-0.055	1.110	1.055	-0.046	0.607	0.561
(-2)A	-1.491	-0.485	-1.976	-0.841	0.879	0.038
(-2)T	-0.517	2.260	1.743	-0.440	0.874	0.434
(-2)C	-0.676	-1.397	-2.073	-0.476	0.324	-0.152
(-2)G	-1.091	4.254	3.163	-0.770	1.729	0.959
(0)C	0.000	0.000	0.000	0.000	0.000	0.000
(+1)A	-5.508	-0.005	-5.513	-5.990	0.586	-5.404
(+1)T	-11.295	0.667	-10.628	-12.118	2.192	-9.926
(+1)C	-7.163	1.154	-6.009	-8.088	2.974	-5.114
(+1)G	-5.035	0.501	-4.534	-5.372	0.229	-5.143
(+2)A	-0.182	0.083	-0.099	-0.181	0.185	0.004
(+2)T	-0.177	-0.191	-0.368	-0.179	-0.035	-0.214
(+2)C	-0.164	0.279	0.115	-0.166	0.649	0.483
(+2)G	-0.163	-0.284	-0.447	-0.159	-0.445	-0.604
(+3)A	-0.024	0.036	0.012	-0.023	0.052	0.029
(+3)T	-0.021	-0.133	-0.154	-0.021	-0.121	-0.142
(+3)C	-0.020	0.119	0.099	-0.020	0.206	0.186
(+3)G	-0.022	-0.195	-0.217	-0.021	-0.253	-0.274
(+4)A	-0.007	0.003	-0.004	-0.006	0.010	0.004
(+4)T	-0.005	-0.091	-0.096	-0.005	-0.095	-0.1
(+4)C	-0.005	0.033	0.028	-0.005	0.065	0.06
(+4)G	-0.006	-0.095	-0.101	-0.006	-0.130	-0.136
(+5)A	-0.003	-0.019	-0.022	-0.003	-0.014	-0.017
(+5)T	-0.002	-0.068	-0.07	-0.002	-0.073	-0.075
(+5)C	-0.002	-0.025	-0.027	-0.002	-0.006	-0.008
(+5)G	-0.003	-0.017	-0.02	-0.002	-0.043	-0.045

Tab. 6 Interaction energy components calculated for 5-methylcytosine in B-DNA. The left part shows values for the substrate model (S), while the right part shows values for the transition state model (TS).

Nucleotide position	Das [kcal/mol]	CAMM [kcal/mol]	MED [kcal/mol]	Das [kcal/mol]	CAMM [kcal/mol]	MED [kcal/mol]
(-5)A	-0.002	0.012	0.01	-0.002	0.015	0.013
(-5)T	-0.001	-0.001	-0.002	-0.001	0.003	0.002
(-5)C	-0.001	-0.002	-0.003	-0.001	0.000	-0.001
(-5)G	-0.002	0.061	0.059	-0.002	0.051	0.049
(-4)A	-0.007	0.031	0.024	-0.007	0.037	0.03
(-4)T	-0.005	-0.000	-0.005	-0.005	0.028	0.023
(-4)C	-0.005	-0.008	-0.013	-0.005	0.018	0.013
(-4)G	-0.008	0.146	0.138	-0.008	0.105	0.097
(-3)A	-0.047	0.094	0.047	-0.038	0.092	0.054
(-3)T	-0.037	-0.074	-0.111	-0.029	0.019	-0.01
(-3)C	-0.035	-0.091	-0.126	-0.028	0.078	0.05
(-3)G	-0.050	0.378	0.328	-0.041	0.181	0.14
(-2)A	-1.268	0.733	-0.535	-0.665	0.293	-0.372
(-2)T	-0.973	-0.357	-1.33	-0.481	-0.492	-0.973
(-2)C	-0.947	-0.580	-1.527	-0.470	0.192	-0.278
(-2)G	-1.269	1.798	0.529	-0.692	0.184	-0.508
(0)C	0.000	0.000	0.000	0.000	0.000	0.000
(+1)A	-4.892	0.363	-4.529	-5.260	0.514	-4.746
(+1)T	-8.047	0.080	-7.967	-8.357	1.114	-7.243
(+1)C	-5.762	2.351	-3.411	-6.117	2.877	-3.24
(+1)G	-4.521	-0.962	-5.483	-4.844	-1.489	-6.333
(+2)A	-0.175	0.127	-0.048	-0.179	0.116	-0.063
(+2)T	-0.114	-0.301	-0.415	-0.118	-0.164	-0.282
(+2)C	-0.114	0.483	0.369	-0.118	0.761	0.643
(+2)G	-0.180	-0.475	-0.655	-0.182	-0.872	-1.054
(+3)A	-0.019	0.008	-0.011	-0.019	0.005	-0.014
(+3)T	-0.013	-0.114	-0.127	-0.013	-0.115	-0.128
(+3)C	-0.012	0.048	0.036	-0.013	0.148	0.135
(+3)G	-0.021	-0.049	-0.07	-0.021	-0.184	-0.205
(+4)A	-0.004	-0.020	-0.024	-0.004	-0.026	-0.03
(+4)T	-0.003	-0.053	-0.056	-0.003	-0.071	-0.074
(+4)C	-0.003	-0.052	-0.055	-0.003	-0.030	-0.033
(+4)G	-0.005	0.060	0.055	-0.005	0.017	0.012
(+5)A	-0.001	-0.020	-0.021	-0.001	-0.026	-0.027
(+5)T	-0.001	-0.031	-0.032	-0.001	-0.044	-0.045
(+5)C	-0.001	-0.069	-0.07	-0.001	-0.071	-0.072
(+5)G	-0.001	0.069	0.068	-0.001	0.061	0.06

Tab. 7 DTSS values for cytosine and 5-methylcytosine in A-DNA and B-DNA models.

Nucleotide position	DTSS C A-DNA [kcal/mol]	DTSS C B-DNA [kcal/mol]	DTSS 5mC A-DNA [kcal/mol]	DTSS 5mC B-DNA [kcal/mol]
(-5)A	0.000	0.001	0.001	0.003
(-5)T	-0.024	0.002	-0.018	0.004
(-5)C	0.045	0.000	0.055	0.002
(-5)G	-0.087	-0.011	-0.095	-0.010
(-4)A	0.028	0.004	0.029	0.006
(-4)T	-0.073	0.021	-0.071	0.028
(-4)C	0.115	0.018	0.132	0.026
(-4)G	-0.200	-0.037	-0.220	-0.041
(-3)A	0.156	0.000	0.162	0.007
(-3)T	-0.262	0.094	-0.271	0.101
(-3)C	0.304	0.144	0.331	0.176
(-3)G	-0.460	-0.171	-0.494	-0.188
(-2)A	1.970	0.249	2.014	0.163
(-2)T	-1.285	0.528	-1.309	0.357
(-2)C	1.910	1.189	1.921	1.249
(-2)G	-2.158	-0.822	-2.204	-1.037
(0)C	0.000	0.000	0.000	0.000
(+1)A	0.318	0.093	0.109	-0.217
(+1)T	1.099	0.900	0.702	0.724
(+1)C	1.035	0.569	0.895	0.171
(+1)G	-0.146	-0.540	-0.609	-0.850
(+2)A	0.131	0.028	0.103	-0.015
(+2)T	0.175	0.136	0.154	0.133
(+2)C	0.402	0.334	0.368	0.274
(+2)G	-0.135	-0.434	-0.157	-0.399
(+3)A	0.028	0.009	0.017	-0.003
(+3)T	0.015	0.006	0.012	-0.001
(+3)C	0.103	0.126	0.087	0.099
(+3)G	-0.055	-0.149	-0.057	-0.135
(+4)A	0.010	-0.005	0.008	-0.006
(+4)T	-0.007	-0.016	-0.004	-0.018
(+4)C	0.036	0.031	0.032	0.022
(+4)G	-0.033	-0.045	-0.035	-0.043
(+5)A	0.005	-0.006	0.005	-0.006
(+5)T	-0.008	-0.015	-0.005	-0.013
(+5)C	0.016	-0.003	0.019	-0.002
(+5)G	-0.022	-0.006	-0.025	-0.008